

How Do Banks Compete?

Evidence from Advertising Videos^{*}

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Abstract

This paper studies how banks compete through a novel lens: the content of advertising videos. Using video embeddings, we identify three competitive dimensions: pricing, service quality, and trust-building emotional appeals. We build a framework connecting advertising content to franchise value and monetary policy transmission. Empirical evidence is consistent with model predictions. Banks with high local market shares compete on service and trust while downplaying pricing. New entrants compete primarily on pricing. Banks lacking pricing or service advantages lean on emotional appeals. Non-price advertising strengthens the deposit franchise, enabling banks to maintain wider spreads during monetary tightening and shaping monetary transmission.

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Understanding how banks compete is central to economics and policy. The intensity and nature of bank competition shape individual banks’ pricing and business strategies, households’ and firms’ access to financial services, financial stability, and the transmission of monetary policy.¹ The existing body of work studies bank competition through the lens of equilibrium outcomes such as prices and quantities.² However, it remains less well documented which specific actions and strategies banks actively use to compete for depositors and borrowers, and to build and defend franchise value. These actions include pricing and, quite importantly, non-pricing dimensions that are often omitted in existing studies.³

This paper studies how banks compete through a novel lens: bank advertising content. Advertising provides a rich yet underutilized window into how banks compete. Advertising videos convey detailed messages about pricing, and more importantly, about services, convenience, trust, and emotional appeal—dimensions of competition that are difficult to observe using standard balance-sheet or pricing data. Advertising is economically important: surveys of Chief Marketing Officers indicate that banks allocate, on average, 9.5% of their annual budgets to advertising, making it a first-order strategic decision rather than a residual expense (Mullainathan, Schwartzstein, and Shleifer, 2008; Bertrand, Karlan, Mullainathan, Shafir, and Zinman, 2010; Gurun, Matvos, and Seru, 2016; Célérier and Tak, 2025).

We use bank advertising content to quantify how banks compete. Our data come from Nielsen Ad Intel and contain 51,349 advertising video clips linked to 2,739 financial institutions, together with airtime schedules that capture the geographic targeting of advertisements across 210 designated market areas (DMAs), spanning 3,098 U.S. counties, from 2004 to 2020. We merge this advertising data with bank balance sheet data from U.S. Call Reports, branch-level deposit data from the FDIC Summary of Deposits, mortgage data from HMDA, and small business lending data from the CRA.

We develop an unsupervised video-embedding method to quantify the content of banks’

¹See, Keeley (1990); Petersen and Rajan (1995); Jayaratne and Strahan (1996); Egan et al. (2017); Carlson et al. (2022); Drechsler et al. (2024); Haddad et al. (2023); Jiang et al. (2024a), and Carletti et al. (2024) for surveys.

²For instance, market concentration and local market power allow some banks to pay lower deposit rates and charge higher loan rates (Rice and Strahan, 2010; Berger and Hannan, 1989; Marquez, 2002; Boyd and De Nicolo, 2005). Moreover, banks with strong deposit franchises can keep deposit rates unusually low when policy rates rise, widening net interest margins and dampening the pass-through of monetary tightening to the real economy (Neumark and Sharpe, 1992; Drechsler, Savov, and Schnabl, 2017; Wang, Whited, Wu, and Xiao, 2022).

³Competition often operates beyond prices, generating heterogeneity in banks’ business models, including asset composition, branch networks, service intensity, and productivity (Egan, Hortaçsu, and Matvos, 2017; Egan, Lewellen, and Sunderam, 2022; Benmelech, Yang, and Zator, 2023; d’Avernas, Eisfeldt, Huang, Stanton, and Wallace, 2023).

television advertisements. Each clip averages 27 seconds and combines visual, verbal, and audio information. Our method represents each video as a high-dimensional embedding that captures this multimodal content, then clusters videos that share similar narratives and scenes. The clusters are chosen to maximize explained variation across videos, requiring no human coding or annotation. The method, though fully unsupervised, recovers economically interpretable categories. Each video projects onto three content types: pricing—interest rates, cashback rewards, and promotional offers; service quality—branch access, mobile banking, and convenience; and trust and emotion—relationship-building, financial advice, and aspirational narratives such as homeownership and small-business success.⁴

Banks’ advertisements exhibit significant diversity in how they compete. 18% of advertisements emphasize their pricing advantages and financial benefits, and 82% on non-pricing information: 29% on customer service and support, and 53% on building trust and emotional connections. A variance decomposition reveals that advertising content is primarily explained by bank, bank×year, and bank×county fixed effects rather than product or region effects. Banks appear to choose distinct competitive positions at the local-market level and sustain them over time.

We develop a parsimonious framework to formalize how banks compete on advertising content and how these choices shape franchise value and monetary policy transmission. Two banks differing in operational reach—branch density, customer support, brand recognition (empirically proxied by local deposit market share)—compete by setting deposit rates and allocating advertising across three content dimensions: pricing, service, and trust. The framework’s central mechanism is that depositor attention to rates is shaped by the content banks broadcast: pricing advertising raises rate awareness, while service and trust advertising directs attention away from rates. Within non-price content, service operates as an informative channel that conveys verifiable quality (such as branch coverage and digital convenience), while trust operates as a persuasive channel that builds attachment independently of underlying quality (Anderson and Renault, 2006; Bagwell, 2007).

In equilibrium, banks with greater operational reach displace pricing content with non-price content—a deposit-market version of the “fat cat” effect (Fudenberg and Tirole, 1984): by drawing depositors toward convenience and away from rates, a high-reach bank softens price competition and sustains a wider deposit spread. Within non-price content, high-reach

⁴This clustering aligns with the marketing literature on advertising content (Resnik and Stern, 1977; Liaukonyte et al., 2015; Guitart and Stremersch, 2021; Tsai and Honka, 2021; Jiang and Kim, 2024) and the finance literature on financial intermediation (d’Avernas et al., 2023; Jiang et al., 2024b; Zhang et al., 2024).

banks tilt toward service, whose returns are amplified by their operational quality, while thinner-franchise banks tilt toward trust, which requires no verifiable backing. Banks lacking both operational quality and brand attachment must instead compete on price. Because television advertising is broadcast to the entire local market, the shift away from pricing content also lowers rate awareness for all depositors in the market, not just the advertiser’s own.

Extending the model to two periods with sleepy depositors (Egan et al., 2025; Lu and Wu, 2026), we trace how content choices shape the transmission of monetary policy. Lower-rate-attentive depositors are slower to leave when banks lag the policy rate, so non-price content lowers deposit betas, and high-reach banks exhibit lower pass-through. The novelty is that depositor composition emerges endogenously from banks’ competitive content choices rather than from monitoring costs, deposit insurance, or individual inattention—a content-based mechanism that complements Drechsler et al. (2017) and Egan et al. (2025). The same general logic extends to the lending market.

Our empirical analysis first examines how banks’ advertising content varies with local market competition. In a panel regression with county-year fixed effects, consistent with our theoretical framework, banks with higher market shares prioritize content on service quality and/or life aspiration over pricing information. Banks with an interquartile increase in mortgage market share are correlated with a 2.5% lower mentions of pricing-related content and a corresponding shift toward service and/or trust-related content. This effect is robust across a wide range of market power and competition measures, including raw county-level market share, the relative ranking of a bank compared to the leader in a region, and the bank-level Herfindahl-Hirschman index (HHI).

We find consistent evidence using alternative empirical designs exploring events that change banks’ competitive environment and market power, such as bank entry into a new market (Egan et al., 2025), bank mergers (Berger et al., 1998; Erel, 2011; Liebersohn, 2024), and the arrival of FinTech competitors (Buchak et al., 2018; Gopal and Schnabl, 2022; Koont, 2023; Erel et al., 2023; Jiang et al., 2024b). When entering a new market through branch openings, these organic entrants, which have little market power, emphasize pricing advantages to attract rate-sensitive customers. They also strengthen trust-building appeals to overcome the entry barrier caused by a lack of trust (Yang, 2025). In contrast, banks entering the market via mergers and acquisitions inherit an existing customer base and shift their advertising efforts toward service quality to retain customers. In response to the expan-

sion of FinTech lenders into local markets, incumbent banks increase their service-focused content, underscoring digital convenience and customer service to counter the technological appeal of their FinTech rivals.

Interestingly, banks compete differently for deposit and loan products. Service quality is emphasized more prominently in deposit product advertisements by banks with high market power, consistent with [d’Avernas et al. \(2023\)](#), who highlight superior liquidity services but low deposit rates in the deposit business of large banks. For loan products, trust-building and emotional appeals are more widely adopted by banks with market power. This aligns well with the recent literature documenting trust being an important factor in households’ financial decision-making ([Guiso et al., 2008](#); [Thakor and Merton, 2024](#)). In line with this, reputation and trust shocks also trigger adjustments to how banks compete. Following the 2016 Wells Fargo scandal, a severe adverse shock to trust in the banking sector, Wells Fargo itself increased its reliance on trust-building by nearly 14 percent, a strategic attempt to repair reputation damage. Meanwhile, other banks in affected counties also increased their trust-focused messaging.

These content choices reflect genuine competitive advantages, not cheap talk, one of our theoretical implications in line with [Anderson and Renault \(2006\)](#). High-rate banks emphasize competitive pricing to attract rate-sensitive depositors; low-rate banks prioritize trust and emotional appeals to retain sticky customers ([Zhang et al., 2024](#)). Banks with high service quality—measured by low consumer complaint ratios—highlight value-added services in their advertising ([Jiang et al., 2024b](#); [Zhang et al., 2024](#)). Content choices thus align with underlying bank characteristics.

Beyond these cross-sectional and quasi-experimental findings, bank competition also interacts closely with the monetary policy environment ([Drechsler et al., 2017](#)). Banks adjust advertising content across the interest rate cycle: when rates rise, advertisements shift toward trust and emotional appeals; during low-rate periods, banks emphasize pricing. In panel regressions following [Drechsler et al. \(2017\)](#) that incorporate both advertising content and deposit market concentration, banks emphasizing service content experience larger deposit spread increases when the Fed funds rate rises, accompanied by larger deposit outflows and sharper balance sheet contraction. These patterns are consistent with non-price content reducing deposit rate sensitivity and amplifying the deposit channel of monetary policy.

The importance of advertising lies in its ability to change demand for banks’ products and their value—we provide supportive evidence in the last part of the paper. To address po-

tential endogeneity concerns for a causal interpretation, we implement a border discontinuity design (Shapiro, 2018, 2020). This strategy leverages the natural geographic discontinuities at the borders of DMAs. Under the identifying assumption that consumers on either side of a DMA border share similar observable characteristics and product choices, variations in exposures to different levels and compositions of advertising due to DMA-specific targeting can help us to identify the effect of advertising. Our design effectively compares the outcomes of the same bank across counties separated by DMA borders, which we interpret as the effects of advertising on consumer demand. We find that advertising content significantly influences consumer demand for deposits, mortgages, and small business loans.

Market power and franchise value are key drivers of bank value (Demsetz et al., 1996; Chen et al., 2022; Egan et al., 2022; Wang et al., 2022; Drechsler et al., 2024), and franchise value is central to monetary policy transmission and interest rate risk (Drechsler et al., 2017, 2021). Dynamic pricing and financial incentives are increasingly studied as tools to compete for depositors (Egan et al., 2025; Benetton et al., 2025). We leverage advertising content to examine how banks compete and sustain franchise value through both pricing and non-pricing messages, with competitive behaviors varying across markets, products, and monetary cycles. An emerging literature documents advertising’s importance across financial markets—mortgages (Gurun et al., 2016; Grundl and Kim, 2019; Kim et al., 2023; Jiang and Kim, 2024), insurance (Shapiro, 2020), credit cards (Bertrand et al., 2010), mutual funds (Roussanov et al., 2021; Hastings et al., 2017), banking (Honka et al., 2017; Célériér and Tak, 2025; Mendes, 2024), and investor trading (Liaukonytė and Žaldokas, 2022)—and broader work emphasizes customer capital accumulated through marketing and branding (Gourio and Rudanko, 2014; He, Mostrom, and Sufi, 2024). We extend this literature by comprehensively examining advertising content in banking and documenting how content choices map to competitive strategy.

This paper also contributes to the burgeoning literature on the use of unstructured data in economics and finance (Gentzkow et al., 2019; Loughran and McDonald, 2020; Hu and Ma, 2025). Closely related to ours, earlier research examines how Fed communication affects monetary policy transmission and financial markets, using audio (Gorodnichenko et al., 2023), text (Shapiro and Wilson, 2022; Gómez-Cram and Grotteria, 2022), and image data (Curti and Kazinnik, 2023). Our paper extends standard approaches, such as word embeddings (Jha et al., 2025), and is among the first to use multimodal models and video embeddings in finance, adding to the literature on AI finance (Eisfeldt and Schubert, 2024).

I. Data

A. Financial Advertising

We obtain advertising data from Nielsen’s Ad Intel dataset, covering national and local television advertisements from 2004 to 2020. The dataset has two components: (i) advertising creative files—video clips of individual advertisements (the “*Creatives*” data); and (ii) airtime schedules tracking each creative’s broadcast across channels, geographic regions, and time slots (the “*Schedule*” data). While prior work has employed schedule data (Shapiro, 2020, 2018; Argente et al., 2025), ours is the first to use the creative video content in academic research.

The creatives data contain 51,349 video clips of financial product advertisements, downloaded from Nielsen’s Ad Intel platform (product codes B120 and B150). A unique video ID links each creative to its schedule records. Nielsen tags each creative with the sponsoring financial institution; we manually verify these tags and use them to link advertisements to the institutional data described below. Section II describes how we process the video content.

The schedule data record each creative’s airtime details. Key variables include air date and time, duration, spending, channel/program, and the designated market area (DMA) in which the advertisement aired. We use these data to measure banks’ advertising strategies, including the geographic targeting of advertisements across markets.

[Insert Table 1 Here.]

Table 1 Panel (a) presents summary statistics of the advertising data. The sample contains 51,349 unique advertisements from 2,739 financial institutions, aired across 210 DMAs. Average advertisement duration is 27 seconds (median: 30), and the average creative remains in circulation for 14 months (median: 4). Most advertisements air on local TV (96%), while 20% also air nationally, with overlaps possible. Conditional on airing, the average number of broadcasts per creative is 1,560 on local TV and 1,340 on national TV. By product category, business banking accounts for 31% of advertisements, personal banking for 24%, mortgages for 17%, and personal credit for 8%. Banks may advertise multiple products across different creatives—for instance, both mortgages and business loans.

B. Banking Data

We merge the advertising data with five major datasets covering bank balance sheets, deposits, mortgage originations, and small business lending: the Federal Reserve’s Call Reports,

Compustat bank fundamentals, the Federal Deposit Insurance Corporation’s (FDIC) Summary of Deposits (SOD), Home Mortgage Disclosure Act (HMDA) data, and Community Reinvestment Act (CRA) small business lending data.

We obtain quarterly bank balance sheet data from U.S. Call Reports via Wharton Research Data Services (WRDS), covering all U.S. commercial banks and their holding companies. We link bank-level figures to branch-level deposit information from FDIC’s SOD, which covers the universe of U.S. bank branches annually from 1993 to 2021. We augment these deposit data with retail deposit rates from Ratewatch, which collects weekly branch-level rates by product and covers 54% of all U.S. branches as of 2013. For publicly listed bank holding companies, we supplement balance sheet data with advertising expenses from Compustat, using the linkage file provided by the Federal Reserve Bank of New York.⁵ We also incorporate deposit betas from Drechsler et al. (2017) and Drechsler et al. (2021), and productivity measures from Egan et al. (2022).

On the lending side, we use HMDA data to measure mortgage originations and local market shares. HMDA covers the near universe of U.S. mortgage applications—both originated and rejected—with detailed loan-level information on borrowers, loans, properties, and lenders. These data include non-bank lenders such as fintech firms, extending coverage beyond traditional banks.

We obtain small business lending data from the CRA dataset. Despite known limitations, CRA data have been widely employed in finance research on bank lending. These data cover business loans with commitment amounts below \$1 million originated by banks with more than \$1 billion in assets. We use them to construct small business loan volumes and local market shares for each financial institution.

C. Sample Construction and Description

We link the banking datasets using standard identifiers (the Federal Reserve’s RSSD ID, the FDIC’s certificate number, and the HMDA’s respondent ID) and merge them with the advertising data via fuzzy matching on institution names.

Table 1 Panel (b) reports summary statistics at the bank-year level. Sample banks hold average deposits of \$15,048 million, with savings APY of 0.62% and time deposit APY of 1.36%, and deposit market HHI of 0.21. Average mortgage originations and small-business

⁵We access the data as of March 2024 at https://www.newyorkfed.org/research/banking_research/crsp-frb, which contains PERMCO-RSSD links from June 30, 1986 to September 30, 2023.

loan volumes are \$2,180 million and \$911 million, respectively.

Table 1 Panel (c) reports bank-county-year-level statistics. The average bank holds \$909 million in deposits per county, representing 31.98% of local deposits. Average savings spread and time deposit spread are 0.70% and -0.21% , respectively. Average mortgage originations and business loans per county are \$19 million and \$4.53 million, accounting for 7.36% and 9.65% of county-level totals.

Which banks advertise on TV? While virtually all banks advertise in some form, TV advertising is concentrated among a subset. In the Internet Appendix (Figure A.1 and Table A.1), we characterize this selection. Banks that are larger, face intense local competition (low HHI), or occupy weaker market positions (lower ROA and market power) are more likely to appear in our sample.

II. Method: Quantifying How Banks Compete Using Advertising

This section describes our key methodological contribution: an empirical framework that quantifies advertising content as a measure of how banks compete. Building on recent work using video as a data input (Hu and Ma, 2025; Huang et al., 2023), our method captures rich information across three dimensions—visual, verbal, and vocal—and projects advertising videos into content categories.

Our key innovation relative to prior work is an unsupervised video-embedding clustering approach that jointly integrates all three information channels, yielding robust and efficient categorization. The approach avoids manual coding, enhances replicability, and applies broadly to other settings requiring video data categorization.

A. An Unsupervised Video Embedding Clustering Method

The method operates in two stages: (1) multimodal embedding, which generates unified vector representations of each video; and (2) hierarchical clustering, which identifies the most salient content dimensions and assigns videos to categories. We outline the method here; the Internet Appendix provides full technical details.

Step 1: High-Dimensional Multimodal Video Embedding We extract information from advertising videos using `OmniEmbed`, an open-source multimodal model built on Alibaba Cloud’s `Qwen2.5-Omni-7B`. `OmniEmbed` jointly processes text, audio, and video inputs and produces a unified embedding—a numerical vector representation—for each advertisement.

In practice, we first separate the audio track and extract representative visual frames from each video. We transcribe the audio using the speech-to-text model **Whisper** (OpenAI), yielding transcripts averaging 53 words per video. The transcript, audio track, and visual frames are then jointly processed by **OmniEmbed**, which generates a 3,584-dimensional embedding for each video. These embeddings encode semantic content and cross-modal interdependencies, integrating all three information channels into a single vector space.

Step 2: Hierarchical Agglomerative Clustering Using Video Embeddings We cluster videos based on the embeddings from Step 1. The guiding principle is that the most salient variance in embedding space reflects the core distinctive features of the videos.

The clustering proceeds in three sub-steps. First, we apply PaCMAP (Pairwise Controlled Manifold Approximation Projection), a dimension reduction method that preserves both global and local structure, to project the 3,584-dimensional embedding vectors (denoted d_v) into a 2-dimensional space $\tilde{d}_v$.

Second, we apply hierarchical agglomerative clustering (HAC) to the projected embeddings $\tilde{d}_v$. HAC constructs a hierarchy of clusters, enabling interpretation at multiple levels of granularity. We measure similarity via Euclidean distance and use Ward’s linkage, which minimizes within-cluster variance at each merge. Because advertisements often span multiple content dimensions, we compute soft cluster memberships using a Student’s t-distribution kernel that converts distances into probabilities, yielding a probability distribution over clusters for each video.

Third, to facilitate economic interpretation, we label clusters using top keywords. We generate natural-language descriptions of each video using **Qwen2.5-Omni-7B** and apply term frequency-inverse document frequency (TF-IDF; Loughran and McDonald, 2011) to identify distinguishing terms. Applying TF-IDF to standardized descriptions rather than raw transcripts isolates distinguishing features while suppressing idiosyncratic word choices and generic banking terms.

Step 3: Quantifying and Interpreting Banks’ Advertising Contents Before clustering on content, we residualize the embeddings to isolate messaging choices from product-level differences. Specifically, we regress each dimension of the 3,584-dimensional embedding on product category fixed effects and retain the residuals, removing variation attributable to the advertised product while preserving heterogeneity in how banks communicate about the

same products.⁶

Applying dimension reduction and clustering (as in Step 2) to the residualized embeddings, we identify three content categories: (1) financial advantage and benefit, (2) service quality, and (3) trust-building and life aspirations. This classification balances comprehensiveness with parsimony, providing categories that are economically meaningful and reliably coded. Figure 1 shows the clustering results (Figure A.2 in the Internet Appendix provides finer detail).

[Insert Figure 1 Here.]

Table 2 summarizes the three broad content themes and their subcategories, with top keywords for each. The “financial information” cluster contains subcategories for interest rates, payment plans, and cashback rewards, identified by keywords such as “rate,” “payment,” and “earning.” The “service quality” cluster includes local service, online service, and daily convenience, characterized by terms like “service,” “online,” “local,” and “community.”⁷

[Insert Table 2 Here.]

These clusters emerge organically from the unsupervised procedure. The categories align with frameworks in marketing for coding advertising content and styles (Liaukonyte et al., 2015; Guitart and Stremersch, 2021; Tsai and Honka, 2021; Jiang and Kim, 2024) and with the finance literature on dimensions of intermediary differentiation and consumer behavior (d’Avernas et al., 2023; Jiang et al., 2024b; Zhang et al., 2024).

Step 4: Aggregate Banks’ Advertising Contents To connect video-level measures to empirical analysis, we aggregate to bank-year, bank-county-year, or other relevant units. For each bank-year, we compute a weighted average of the residualized embeddings, where weights reflect each advertisement’s normalized airtime frequency. We then apply dimension reduction and clustering (Steps 2–3) to these aggregated embeddings to obtain content category distributions at the bank-year level. This procedure extends naturally to finer aggregations—for instance, by product side (liability vs. asset) or by geography—allowing us to examine how banks tailor their messaging across markets and product lines.

⁶Product categories are identified as a preliminary step of our algorithm; see Internet Appendix A.2.2.

⁷Table A.3 compares our clustering with alternative ML methods, showing strong consistency.

B. Examples of Banks' Advertising Content and Our Method

In this section, we present examples of outcomes from these methods.⁸

Example of Financial Information The financial information category intends to capture the financial benefits of the product, including interest rates, payment plans, and cashback rewards.

“You’ve got more options than you think. And our HARP loan lets you refinance at a much lower rate, even if you owe more than your home is worth. Lower your monthly payment and free up some cash flow. You can qualify right over the phone. Today’s refinance rate is 3.5% APR.”

Example of Service Quality The service quality category includes information on local and community service, online and mobile applications, and daily convenience.

“You’re going to love our new mobile app. Bank from your computer, tablet, or phone with the same look and feel and only one user ID and password. Pay quickly by taking a picture of a paper bill or pay anyone anywhere. It’s all so easy with a mobile app.”

Example of Emotional Appeal and Trust Building Our last broad bank advertising content is the emotional appeal and trust building. For emotional appeal, we focus on the small business support, home and future, college and education, and sports sponsorship.

“There is nothing like a home in Maine and a bank that makes you feel at home. We have money to lend for dreams like yours with real people you can talk to and smart solutions right here. We’re strong and sound with the mortgage you need for the home of your dreams. Catch a great rate at VAT Savings when you’re here, your home. Equal Housing Lender, member FDIC. There’s a confidence in there.”

For trust building, we focus on relationship building, including brand promotion and fostering trust in the home and community, as well as professional imagery that showcases expertise and wealth management.

⁸We provide more related examples of advertising content in Appendix [A.2.4](#).

“We are here to talk about the recent changes in the government-insured reverse mortgage program. Making it safer for homeowners. The benefits of a reverse mortgage are tremendous. Giving seniors the financial freedom and security while allowing seniors to stay in their home. While turning their home equity into tax-free monthly income. So are you 62 or older, on your own, and need a steady income flow each month? Then call my friend”

C. Advertising Content Distribution and Variance Decomposition

Figure 2 shows the distribution of content across creatives.⁹ Emotional appeals and trust-building dominate (53%), followed by service quality (29%) and pricing/financial information (18%). The dominance of emotional and trust-based messaging is notable: [Bertrand et al. \(2010\)](#) show that concrete financial information can reduce consumer demand, while content appealing “peripherally” to intuition and emotions significantly increases it. Banks appear to have internalized this principle. The weight on trust-building in particular aligns with [Thakor and Merton \(2024\)](#) and [Yang \(2025\)](#), who theorize and document the importance of trust in bank entry and lending relationships.¹⁰

[Insert Figure 2 Here.]

Internet Appendix Figure A.3 provides finer categorizations. Within financial advantages, interest rate content is most prevalent (8%), followed by payment plans (6%) and cashback rewards (4%). Within service quality, local service leads (14%), with online service (8%) and daily convenience (7%) following. Within emotional appeals and trust-building, relationship building is the largest subcategory (23%), with emotional imagery (15%) and professional imagery (15%) equally represented.

We next decompose the variance in the content banks deliver in their advertisements. We do so using a bank-county-product-year panel and regressing banks’ content shares on several sets of fixed effects. To assess the relative importance of each set, we employ a Shapley-Owen decomposition, which provides a principled way to attribute explanatory power in a regression with multiple, potentially overlapping sources of variation. The procedure considers all possible orderings of introducing the fixed effects into the regression and averages their incremental contributions to model fit (R^2), ensuring that no set of fixed effects receives undue

⁹Results are similar when weighted by airtime.

¹⁰Content distribution may vary across media channels ([Abernethy and Franke, 1996](#)) and thus may not generalize beyond TV advertising.

priority simply because of the order in which it enters. This approach is particularly useful in our setting, where the explanatory factors—bank, geography, product, and time—are likely correlated.

Figure 3 presents a clear picture: the bulk of observed differences in banks’ advertising content are attributable to bank-related dimensions, indicating that banks make persistent, strategic choices about how to compete through advertising. The baseline begins with product $\times$ year fixed effects. Only approximately 2% of advertising variation can be explained by banks using different content across different financial products. Although it is intuitive that different products may require slightly different advertising strategies, this force plays a limited role. Adding county $\times$ year fixed effects increases R^2 by about 3.6%, suggesting that differences across local markets—including local concentration and economic conditions—are not the primary drivers of banks’ advertising decisions.

[Insert Figure 3 Here.]

We then examine time-invariant and time-varying bank heterogeneity. Bank fixed effects alone increase R^2 by around 5.8%. There is a substantially larger increase when time-varying bank effects are added: bank $\times$ year fixed effects produce a 31% increase in explained variation for content distributions, indicating that banks’ advertising strategies evolve meaningfully over time. Finally, bank $\times$ county fixed effects, which capture the matching between banks and local markets, explain an additional 12.8% of content variation. Together, these bank-related dimensions account for the bulk of explained variation, confirming that advertising content is shaped primarily by bank-level strategic choices rather than by product or geographic characteristics.

III. A Conceptual Framework

This section develops a parsimonious framework in which banks allocate advertising across three content dimensions—pricing, service, and trust—each shaping deposit demand through a distinct mechanism. Pricing content raises depositors’ rate awareness; service content conveys verifiable quality; trust content builds emotional attachment that operates independently of quality. Banks’ endogenous allocation across these three margins determines the rate sensitivity of their depositor base and, through it, their market power and the transmission of monetary policy. We present a duopoly model, derive equilibrium predictions, and discuss implications for deposit rate pass-through. All proofs are in the Online Appendix.

A. Model Setup

Environment Two banks $j \in \{H, L\}$ compete for a continuum of depositors (mass 1). Bank H has superior service quality q_H —we interpret this as a bank with a denser branch network or better customer support. Bank L has $q_L < q_H$ —think of a smaller bank with local branch coverage or limited customer service. These quality differences are empirically correlated with local market share: banks with larger share tend to have denser branch networks and more established reputations in the local area, motivating our use of local share as the empirical proxy for market power in Section IV. Both banks invest deposits at the risk-free market rate r_M (taken as exogenous) and compete by setting deposit rates r_j . Following Drechsler et al. (2017), we define the deposit spread as $\rho_j \equiv r_M - r_j$, the bank’s margin per dollar of deposits.

Advertising Content and Depositor Attention Each bank allocates advertising across three dimensions: $a_j^P + a_j^S + a_j^T = 1$. Pricing content a_j^P communicates rate information. Service content a_j^S highlights verifiable attributes—branch amenities, digital offerings—that convey the bank’s underlying product quality. Trust content a_j^T builds emotional attachment and brand loyalty through appeals that operate independently of verifiable quality signaling. Such a decomposition separates the “match” category in Anderson and Renault (2006) into an informative channel (service) and a persuasive channel (trust), following the information-vs.-persuasion framework of Bagwell (2007).

Depositors differ in their attention to interest rates. Motivated by the recent evidence on sleepy depositors,¹¹ we model the effective rate awareness of bank j ’s depositors as a function of market-wide advertising on pricing:

$$\bar{\alpha}_j = \alpha(a_j^P, a_{-j}^P), \quad \text{subject to} \quad \frac{\partial \alpha}{\partial a_j^P} > \frac{\partial \alpha}{\partial a_{-j}^P} > 0. \quad (1)$$

where the function $\alpha(\cdot)$ is smooth and strictly positive.¹² These conditions have a natural economic interpretation. Since television advertising is broadcast to the entire DMA, pricing content from any bank raises rate awareness for all depositors in the local market. The own-bank effect is larger because own pricing content both injects rate information into the

¹¹Egan et al. (2025) document that depositors are “sleepy”: they infrequently monitor rates and rarely switch banks. Using transaction-level data on millions of U.S. depositors, Lu and Wu (2026) show that inattention varies widely across depositors and that more inattentive depositors adjust their balances less in response to monetary policy changes.

¹²We provide more discussion on the effective rate awareness and the function $\alpha(\cdot)$ in Appendix A.3.1.

market and directly orients depositors toward rate comparisons when evaluating that bank, while the rival's pricing content operates through altering the market-level awareness. As such, non-price advertising has a spillover effect: a bank not only puts its own depositors to sleep, but all depositors in the market.

Depositors and Banks We model bank j 's non-price advertising consumed by a depositor as a CES bundle:

$$A_j = [(q_j a_j^S)^{1-1/\eta} + (\xi a_j^T)^{1-1/\eta}]^{\eta/(\eta-1)}, \quad (2)$$

where $\eta > 1$ is the constant elasticity of substitution across service and trust advertising content. The parameter q_j governs the efficiency of service content, which, in turn, affects the return to advertising on service content. That is, high-quality banks benefit more from service advertising and therefore tilt their advertising to service content. $\xi > 0$ is the scaling factor for trust content. We take ξ as common across banks because trust operates through emotional appeal rather than verifiable bank-specific quality. The CES bundling of non-price advertising exhibits diminishing marginal returns within each content dimension: the first advertisement on service generates the highest value.

The mean indirect utility depositors derives from bank j is given by:¹³

$$u_j = -\bar{\alpha}_j \rho_j + A_j, \quad (3)$$

where $\bar{\alpha}_j > 0$ is the bank-specific rate awareness as defined above. Depositors dislike high spreads ($-\bar{\alpha}_j \rho_j < 0$) and derive positive utility from consuming bank non-price advertising ($A_j > 0$). Adding i.i.d. type-1 extreme value taste shocks yields logit market share as

$$D_j = \exp(u_j) / [\exp(u_H) + \exp(u_L) + 1], \quad (4)$$

where the “+1” represents the outside option (e.g., money market funds) with utility $u_0 = 0$. Bank j 's profit Π_j is determined by its deposit spread and market share $\Pi_j = \rho_j D_j$.

¹³Following Egan et al. (2025), we write utility in terms of the deposit spread $\rho_j = r_M - r_j$ rather than the deposit rate r_j . This reformulation implicitly treats the outside option as paying the market rate and introduces a bank-specific shift of $-\bar{\alpha}_j r_M$ in each inside utility. Because this shift is smaller (less negative) for low- $\bar{\alpha}_j$ banks, it reinforces rather than overturns the equilibrium ordering $\rho_H^* > \rho_L^*$ derived below. The spread formulation is convenient because ρ_j directly measures the bank's margin per dollar of deposits.

B. Equilibrium Predictions

A first question Bank H faces is whether it can leverage its quality advantage to compete on non-price dimensions. If Bank H emphasizes rates, it raises rate awareness, forcing rate parity with Bank L and eroding Bank H 's quality premium. Alternatively, by displacing pricing content with non-price advertising, Bank H can reduce its depositors' rate sensitivity and sustain a wide spread. The following proposition characterizes when this non-price advertising equilibrium exists.

Proposition 1 (Content-Driven Market Power). *If quality differentiation is sufficiently large ($q_H/q_L > \theta^*$ for a threshold $\theta^* > 1$), then in every equilibrium: (i) Bank H (high quality) chooses low pricing content and high non-price content, while Bank L chooses high pricing content ($a_H^{P*} < a_L^{P*}$); (ii) within non-price content, the service share $a_j^{S*}/(a_j^{S*} + a_j^{T*})$ is increasing in bank quality; and (iii) equilibrium spread is larger for Bank H : $\rho_H^* > \rho_L^*$.*

Proof in Appendix A.3.2.

Bank H 's quality advantage makes non-price advertising—especially service content—credible and strategically powerful. When Bank H advertises service, it raises its non-price advertising value A_H , attracting depositors; simultaneously, the displaced pricing content reduces $\bar{\alpha}_H$, lowering rate awareness and widening the deposit spread. Bank L lacks such a quality credibility, and instead raises rate awareness through more pricing advertisements, competing on the price dimension where it has a comparative advantage. Bank H avoids pricing competition because making rates salient would raise sensitivity $\bar{\alpha}_H$, forcing a narrower spread and collapsing the quality-driven demand advantage. This displacement of competition to non-price dimensions is the deposit-market analogue of the “fat cat” effect (Fudenberg and Tirole, 1984): a firm relaxes price competition by investing in brand-building that makes its consumers less price-elastic.

Proposition 2 (Non-Price Content and Spreads). *Non-price content reduces bank-specific rate sensitivity $\bar{\alpha}_j$ and increases equilibrium spreads. For high-quality banks, the spread increase is larger because their high quality amplifies the return to service advertising—the informative channel is more effective when quality is verifiable.*

Proof in Appendix A.3.3.

The mechanism operates through the rate-awareness channel: each unit of bank j 's pricing content displaced by non-price content reduces $\bar{\alpha}_j$, directly widening the markup

$\rho_j^* = 1/[\bar{\alpha}_j(1 - D_j)]$. This effect is symmetric across service and trust content—shifting to either type reduces pricing content equally. Bank quality amplifies this effect: since Bank H has lower equilibrium $\bar{\alpha}_H$ (from Proposition 1 (i)), the marginal spread-widening effect is larger, creating a self-reinforcing feedback. Across banks, an additional demand channel operates through the level comparison: Bank H ’s higher quality raises its demand share D_H , further widening the spread through $(1 - D_j)$ in the denominator. Within non-price content, the service share is increasing in quality (Proposition 1 (ii)), so informative advertising is the primary competitive tool for high-quality banks.

Deposit Rate Pass-Through We extend the model to two periods to trace how advertising content shapes deposit rate pass-through. In Period 1, banks choose advertising content and deposit rates, establishing their depositor base composition. In Period 2, the policy rate rises by $\Delta r > 0$ and banks re-optimize deposit rates with pre-committed advertising content. Depositors who remain asleep do not observe the rate change and stay at their current bank, while awake depositors re-optimize. This depositor stickiness generates heterogeneous pass-through.

Define deposit beta as $\beta_j \equiv \partial r_j^* / \partial r_M$, the fraction of a policy rate increase that bank j passes through. The baseline frictionless logit model implies full pass-through ($\beta_j = 1$ for all banks). However, frictions such as imperfect outside-option tracking, depositor search costs, and deposit insurance (Drechsler et al., 2017) can generate incomplete pass-through. To account for these potential frictions, we adopt the reduced-form assumption that $\beta_j = f(\bar{\alpha}_j)$ with $f' > 0$: banks with more rate-attentive depositors face faster outflows when lagging the market rate and pass through more quickly.¹⁴

Since $\bar{\alpha}_j$ is decreasing in own non-price content (Proposition 2), the deposit beta is also decreasing in non-price content: $\partial \beta_j / \partial a_j^S, \partial \beta_j / \partial a_j^T < 0$. Banks emphasizing non-price content have sleepier depositor bases and lower pass-through. Moreover, $\bar{\alpha}_H < \bar{\alpha}_L$ (since Bank H does more non-price content), so $\beta_H < \beta_L$: high-quality banks exhibit lower deposit betas. This offers a strategic, content-driven explanation for why some banks accumulate predominantly “sleepy” depositors (Egan et al., 2025), extending beyond traditional explanations based on monitoring costs or deposit insurance (Drechsler et al., 2017). Whereas Drechsler et al. (2017) treat depositor composition as exogenous, our model shows that it emerges endogenously from banks’ advertising strategies. Similarly, while Lu and Wu (2026)

¹⁴A tractable parametrization is $\beta_j \approx \bar{\alpha}_j / (\bar{\alpha}_j + \phi)$ with $\phi > 0$. The sign predictions depend only on $f' > 0$. Appendix A.3.4 provides the formal derivation.

view depositor inattention as an individual characteristic, our framework attributes it to banks’ strategic content decisions.

Although developed for deposit markets, the core mechanism extends to lending: non-price advertising can reduce borrowers’ rate sensitivity, enabling wider loan spreads. A key difference is that loan quality is harder for borrowers to assess *ex ante*, so the informative (service) channel is weaker on the asset side. This suggests trust content plays a larger role in loan advertising, a pattern we examine via our border design in Section C.

IV. How Do Banks Compete Using Advertising Content?

This section, guided by our theoretical framework, examines how banks compete through advertising content. We study three sets of predictions from the model: (i) high-market-share banks (proxying for high-quality Bank H) choose less pricing content and more non-price content, with service share increasing in quality (Proposition 1); (ii) content choices depend on entry mode—organic entrants, lacking service reputation, should rely on pricing content, while M&A acquirers, inheriting sticky depositor bases, should shift toward service content; and (iii) banks’ content choices should align with their actual competitive advantages, with the informative (service) channel dominating for high-quality banks (Proposition 2).

A. Local Market Power and Competition Behaviors

We first examine how advertising content varies with local market power in a bank-county-year panel. We estimate the following model in a bank-county-year panel:

$$\text{AdContent}_{b,c,t} = \beta \cdot D_{b,c,t-1} + \eta_{c,t} + \epsilon_{b,c,t}. \quad (5)$$

The dependent variable is the share of content categories—pricing, service quality, and trust-building/emotional appeals—in bank b ’s advertising videos aired in county c in year t . These categories map to the model’s three content dimensions: pricing content p_j , service content s_j , and trust content τ_j .

The key independent variables, denoted D , capture bank characteristics at the b - c - t level, including local market power (county-level market shares), deposit rate-setting behavior, and service quality. We analyze deposits and lending (mortgages and business loans) separately where relevant. County-year fixed effects $\eta_{c,t}$ absorb time-varying local demand, demographics, and competitive conditions, so the specification compares advertising strategies across

banks with different characteristics within each county-year. Standard errors are clustered at the county level.

We proxy for local market power using three measures constructed at the bank–county–year level. The first is the bank’s local market share in a given county and product market (i.e., separately for deposit and loan products), which captures its absolute local presence. Deposit and loan shares are calculated as each bank’s share of county-year deposits and HMDA mortgage originations, respectively. The second measure scales this share by the market share of the bank with the largest share in the same county–product market, measuring the bank’s position relative to the local leader. It reflects whether the bank is the dominant institution versus a smaller competitor. The last measure captures the competitiveness of the bank’s operating footprint using a bank-level Herfindahl–Hirschman index (HHI) measure (Berger and Hannan, 1989). Following Drechsler et al. (2017), we compute county-level HHI in each product market and take a product-weighted average across the counties where the bank operates. Banks that concentrate their activity in more concentrated local markets are classified as having greater market power.

[Insert Table 3 Here.]

Table 3 provides evidence consistent with Proposition 1(i): across all three market power proxies, high-market-share banks allocate significantly less advertising to pricing content. This pattern holds across products. A 10-percentage-point increase in market share (roughly interquartile) reduces mortgage pricing content by 0.368 percentage points—2.5% of the 14.51% mean—with smaller effects for deposits.¹⁵

High-market-share banks compete on non-price dimensions—service quality and trust-building—consistent with Proposition 1. In deposit advertising, they emphasize service quality 1.12 percentage points more per interquartile increase in market share (3.1% of the 36.16% mean). The informative (service) channel appears particularly valuable for high-quality banks: service advertising credibly conveys their quality advantage and redirects depositor attention from rates to quality attributes. This aligns with d’Avernas et al. (2023), who document that large-bank depositors disproportionately value liquidity services and exhibit lower rate sensitivity.

An important deposit–loan asymmetry emerges. While high-market-share banks emphasize service quality on the liability side, they rely more heavily on trust-building and

¹⁵Appendix Table A.5 compares banks in the top 20th percentile of market share against the bottom 80th percentile in deposit and mortgage markets separately, with consistent results.

life-inspirational content for loan products—success stories of small businesses, homeownership narratives. In mortgage advertising, a 10-percentage-point increase in market share raises life-inspiration content by 0.6 percentage points (1.2% of the sample mean). One interpretation, guided by the framework’s lending extension, is that borrowers cannot assess loan quality *ex ante* (unlike depositors, who can verify branch amenities and service), so the informative (service) channel is weaker on the asset side. Trust content, which reduces rate sensitivity regardless of verifiable quality, plays a correspondingly larger role on loans. That said, the pattern may also reflect that loan products inherently lend themselves to aspirational messaging.

These results are consistent with Propositions 1 and 2: high-market-power banks compete via non-price content, with service dominating on the liability side and trust on the asset side. By redirecting depositor attention from rates to quality and brand attributes, non-price advertising reduces effective rate sensitivity $\bar{\alpha}_j$, enabling wider equilibrium spreads. This content-driven mechanism complements switching-cost and deposit-insurance explanations for why high-market-share banks sustain less favorable deposit rates.

B. *Alternative Designs: Bank Entry and Shocks to Trust*

We complement the cross-sectional analysis with quasi-experimental designs exploiting shocks that shift content for reasons tied to entry mode or trust destruction rather than endogenous market structure. We examine three settings: bank entry (organic vs. M&A), FinTech competition, and the Wells Fargo scandal.

We begin with market entry, a setting that allows us to examine Proposition 1’s prediction that content depends on bank type (Egan et al., 2025). Organic entrants, lacking established service reputation (low q_j), should emphasize pricing; M&A acquirers, inheriting customer relationships and quality, should shift toward service.¹⁶ We estimate:

$$\text{AdContent}_{b,c,t} = \beta \cdot \text{Post(Entry)}_{b,c,t} + \eta_b + \eta_{c,t} + \epsilon_{b,c,t}, \quad (6)$$

where $\text{Post(Entry)}_{b,c,t}$ is a dummy variable that takes the value of 1 if year t falls within the three-year window after bank b ’s entry into county c , and the value of 0 otherwise. Bank fixed effects η_b absorb bank time-invariant heterogeneity. County-year fixed effects $\eta_{c,t}$ absorb local economic shocks. Panel (a) of Table 4 presents the results.

¹⁶M&A data downloaded from <https://www.ffiec.gov/npw/FinancialReport/DataDownload>.

[Insert Table 4 Here.]

Entry Through New Branch. For organic entry via branch expansion (columns 1–3), entrants allocate significantly more to pricing and trust content relative to incumbents in the same county-year. This pattern is consistent with Proposition 1: lacking established service reputation, new entrants tend to compete on pricing where they are credible. They offer high deposit rates and low mortgage rates to attract rate-sensitive depositors, exactly as predicted for Bank L in Proposition 1. The concurrent increase in trust content reflects entrants’ need to overcome switching costs and depositor unfamiliarity, consistent with trust serving as an entry barrier (Thakor and Merton, 2024; Yang, 2025).

Entry Through M&A. Banks entering via acquisition inherit established customer bases and service reputations—analogueous to Bank H in the model. Under Proposition 1, these banks should emphasize non-price content, with service share increasing in their inherited quality advantage.

Columns (4)–(6) of Panel (a) document this pattern. Following M&A entry, acquiring banks increase service-content weight by 7.83 percentage points (21% of the 36.61% mean), consistent with Table 3. M&A acquirers inherit sticky depositors and can immediately pivot from price competition to service emphasis, leveraging their inherited quality advantage. This also aligns with Erel (2011), who find that merged banks leverage increased operating efficiency to enhance service supply.

FinTech Entry Into Local Markets We examine how incumbent banks respond to FinTech entry, which exogenously intensifies competition in local lending markets (Buchak et al., 2018; Gopal and Schnabl, 2022; Koont, 2023; Jiang et al., 2024b).

To estimate changes in incumbent banks’ advertising content after FinTech entry, we estimate the following within-bank specification.

$$\text{AdContent}_{b,c,t} = \beta \cdot \text{Post}(\text{FinTech Entry})_{c,t} + \eta_c + \eta_{b,t} + \epsilon_{b,c,t}, \quad (7)$$

$\text{Post}(\text{FinTech Entry})_{c,t}$ is a dummy variable that takes the value of 1 if year t falls within the three-year window post the FinTech’s entry into county c , and the value of 0 otherwise. We also consider an intensive measure of FinTech entry by using the mortgage market share of FinTech companies in county c and year t . We consider the set of FinTech-classified

lenders following [Fuster et al. \(2019\)](#). We add county fixed effects η_c and bank-year fixed effects $\eta_{b,t}$, which control for regional time invariant heterogeneity and the bank’s lending opportunities. As such, we compare competition strategies of the same banks, but across branches in different counties, that may or may not be affected by FinTech entry.

Panel (b) shows that incumbent banks respond by increasing service-quality advertising (digital banking, mobile access, quick approval). Banks facing a quality threat shift toward the informative (service) channel—highlighting verifiable advantages like branch networks and relationship banking—rather than relying on persuasive (trust) appeals, consistent with Proposition 2.

Wells Fargo Scandal We analyze the 2016 Wells Fargo scandal as a distinct trust shock. Unlike the competition shocks above, this event exogenously destroys depositor trust in a specific bank. In 2016, Wells Fargo disclosed years of unauthorized account creation, triggering widespread consumer skepticism. Though fraudulent activity began in 2005, the public revelation was sudden and plausibly orthogonal to industry fundamentals ([Yang, 2025](#)).

We first focus on a sample of Wells Fargo only and examine how Wells Fargo’s advertising content changes after the scandal. Since there is only one bank (i.e., Wells Fargo) in this sample, the panel collapses to the county (c)-year(t) level. We use the specification below to estimate the effect of the scandal on Wells Fargo’s content choice.

$$\text{AdContent}_{c,t} = \beta \cdot \text{Post(Scandal)}_t + \eta_c + \epsilon_{c,t}, \quad (8)$$

where $\text{AdContent}_{c,t}$ is Wells Fargo’s advertising content distribution in county c in year t . Post(Scandal)_t is a dummy variable that takes the value of 1 if year t falls within the five-year window from 2016 to 2020, and the value of 0 otherwise. η_c are county fixed effects to absorb time-invariant local market heterogeneity.

Columns (1)–(3) in Panel (c) show that post-scandal, Wells Fargo sharply increased trust-building and emotional content by 7.7 percentage points (14% of the 53.37% mean), consistent with campaigns like “Re-Established” (“it’s a new day at Wells Fargo”).¹⁷ This response is consistent with the framework: when the scandal destroyed Wells Fargo’s trust capital, the bank adjusted its content mix by increasing persuasive (trust) content τ_j , which

¹⁷See <https://www.latimes.com/business/la-fi-wells-fargo-ad-campaign-20180509-story.html>.

the theory predicts reduces rate sensitivity and could facilitate rebuilding depositor loyalty. The episode suggests advertising content may serve as a reputational repair tool.

We also examine spillover effects on competitors in counties with Wells Fargo branches, estimating:

$$\text{AdContent}_{b,c,t} = \beta \cdot \text{Post(Scandal)}_t \times 1(\text{Wells Fargo Presence})_c + \eta_c + \eta_{b,t} + \epsilon_{b,c,t}, \quad (9)$$

where $\text{AdContent}_{b,c,t}$ is bank b 's advertising content distribution in county c in year t . Post(Scandal)_t is a dummy variable that takes the value of 1 if the year t falls within the five-year window from 2016 to 2020, and the value of 0 otherwise. $1(\text{Wells Fargo Presence})_c$ is a dummy variable which takes the value of 1 if county c has at least one Wells Fargo branch in 2016 or the value of 0 otherwise. η_c are county fixed effects. $\eta_{b,t}$ are bank-year fixed effects. By including the set of bank-year fixed effects, we are comparing advertising content across branches of the same bank, but in different counties, with or without Wells Fargo's physical presence.

Columns (4)–(6) show that competitors in Wells Fargo counties also increase trust-building content, though by less than Wells Fargo. This spillover suggests that trust shocks affect the optimal content mix of all banks in the local market, not just the focal institution.

C. Highlighting Strengths, Substituting Toward Trust

Our framework also implies that banks with genuine advantages in pricing or service quality should highlight these verifiable strengths through informative advertising, leveraging their actual competitive edge. Banks lacking hard advantages in these dimensions should strategically shift toward trust-building and emotional appeals—substituting persuasive content where informative content would not be credible. We examine this mechanism through the lens of the informative versus persuasive advertising distinction (Anderson and Renault, 2006; Bagwell, 2007).

We investigate how financial advertising strategies differ across banks with varying pricing and service quality, and examine whether their advertising content aligns with their pricing advantages and service quality. We first focus on core retail deposit products, including \$10,000 12-month CDs, \$25,000 money market deposit accounts, and interest checking accounts with less than \$2,500, using the data from RateWatch. These products represent the primary types of retail deposits in banks' core offerings. To incorporate information from

all three retail deposit rates, we use a weighted rank method as in [Zhang et al. \(2024\)](#). We first rank all banks based on the three deposit rates in each year and average these ranks. We define high-rate banks as those with average rankings above the 80th percentile within the same county for a specific year, and low-rate banks as those with average rankings below the 80th percentile.

[Insert Table 5 Here.]

High-rate banks emphasize pricing in deposit advertising: 3.13 percentage points more than low-rate banks (23% of the 13.6% mean), consistent with informative advertising—banks with genuine rate advantages make them salient. Low-rate banks compensate by allocating 2.22 percentage points more to service content (6.2% of the mean), redirecting depositor attention from their pricing disadvantage toward quality attributes. Proposition 2 explains this pattern: banks lacking pricing advantages use non-price content to reduce depositors’ effective rate sensitivity.

We next examine whether high-service-quality banks employ informative advertising. We measure service quality using the complaint rate (complaints per dollar of deposits) from the CFPB Consumer Complaint Database; banks with fewer complaints are classified as higher quality ([Egan et al., 2017, 2022](#)).

High-service-quality banks allocate 2.45 percentage points more to service content (6.7% of the mean), consistent with Proposition 2: when quality is verifiable, the informative (service) channel dominates because it amplifies the bank’s genuine advantage (the $\alpha_2 \tilde{q}_j$ mechanism). Banks with concrete service advantages advertise them, attracting depositors who value quality over rate minimization.¹⁸

A revealing cross-cutting pattern: high-rate and high-service-quality banks allocate *less* to trust-building content. Proposition 2 provides a rationale: when a bank has verifiable competitive advantages, the theory predicts the informative channel dominates, making persuasive content less central to the competitive strategy.

These patterns reveal a nuanced advertising strategy that blends informative and persuasive elements. Banks with genuine advantages—competitive rates or superior service—highlight them, consistent with informative advertising. Banks lacking advantages in one dimension strategically reallocate toward other dimensions rather than highlighting their

¹⁸Appendix Table A.4 measures service quality using branch network density following [Egan et al. \(2017\)](#) and [Zhang et al. \(2024\)](#), with consistent results.

weaknesses. This selective emphasis reflects the equilibrium content allocation predicted by Propositions 1 and 2: each bank optimally emphasizes the dimension in which it holds a comparative advantage. Where genuine advantage exists, banks foreground it; where it does not, they foreground trust. This yields a bifurcated advertising market: informative content among high-quality banks, persuasive content among the rest.

V. Advertising, Franchise Value, and Monetary Policy

A key feature of bank competition and market power is its deep connection with monetary policy through the bank deposit channel (Drechsler et al., 2017). We now examine whether advertising content serves as a mechanism through which banks build franchise value and exercise market power, with consequences for how monetary policy transmits to deposit pricing and credit supply.

Our theoretical framework predicts that non-price content—service (informative) and trust (persuasive)—enables higher deposit franchise value, manifested in wider deposit spreads and lower deposit rate sensitivity to Fed funds rate changes, with implications for the strength of the deposit channel. This section proceeds in three steps. First, we link advertising content to bank-level measures of market power and franchise value—deposit productivity, deposit spreads, deposit betas, and Tobin’s q . Second, following Drechsler et al. (2017), we examine how advertising content shapes banks’ heterogeneous responses to monetary policy shocks at the bank-county-year level, controlling for deposit market concentration. Third, we implement a border-discontinuity design (Shapiro, 2018, 2020) that leverages geographic discontinuities in advertising exposure at DMA borders to identify the causal effects of advertising content on demand for deposits, mortgages, and small-business loans.

A. Advertising Content and Bank Value

We first examine whether advertising content is associated with standard measures of deposit franchise value: deposit productivity, deposit spreads (the markup banks earn over the Fed funds rate), deposit betas (rate pass-through sensitivity), and Tobin’s q . We use the bank holding company (b)-quarter (t) panel and estimate the following specification:

$$Y_{b,t} = \beta \cdot \text{AdContent}_{b,t-1} + \gamma X_{b,t-1} + \eta_t + \epsilon_{b,t}, \quad (10)$$

where $Y_{b,t}$ is a set of bank value measures such as productivity, deposit spread, and Tobin’s q of bank b in quarter t . $\text{AdContent}_{b,t-1}$ is the advertising content measures: pricing, service, emotion/trust. Since these three measures sum to 100% by definition, we omit the emotion/trust (persuasive) category to avoid collinearity. Estimated coefficients on pricing and service should therefore be interpreted relative to emotion/trust-focused banks. $X_{b,t-1}$ is a list of lagged bank controls, including the log of the bank’s total assets, the bank’s Z-Score (the bank’s ROA plus its equity ratio divided by its standard deviation of ROA), the bank’s ROA, the bank’s equity to assets ratio, the bank’s deposits to assets ratio and the bank’s deposit and mortgage market concentration HHI, following [Gelman et al. \(2023\)](#) and [Egan et al. \(2022\)](#); η_t are time fixed effects. We use a sample with non-missing and positive advertising expenses. Standard errors are clustered at the bank level using wild bootstrap. This specification leverages cross-sectional variation in banks, since some measures, such as bank productivity, are inherently cross-sectional. To mitigate reverse causality concerns, we lag all advertising measures by one period relative to the dependent variables.

[Insert Table 6 Here.]

Table 6 reports the results.¹⁹ Column (1) shows that banks emphasizing pricing in their advertising display higher deposit productivity, echoing [Egan et al. \(2022\)](#), who emphasize rate-setting technology as an important source of variation in bank productivity. Column (2) reveals that banks with greater emphasis on emotion and trust achieve higher asset productivity, as indicated by negative coefficients on the pricing and service categories. This pattern mirrors our earlier evidence in Table 3 column (6): banks emphasizing the persuasive (trust) channel are more efficient in lending and achieve higher asset productivity. This finding also accords with [Thakor and Merton \(2024\)](#), who demonstrate that borrowers’ trust is crucial for banks to maintain business advantages in lending.

Columns (3) to (5) examine bank market power and franchise value. Following [Drechsler et al. \(2017\)](#) and [Drechsler et al. \(2021\)](#), we use the deposit spread and the deposit beta to measure market power. The deposit spread is the difference between the Fed funds rate and the bank deposit rate—a “markup” that banks charge on deposits. The deposit rate is computed as quarterly total domestic deposit expense divided by quarterly total domestic deposits and then annualized. Column (3) shows that banks emphasizing the informative (service) channel maintain significantly higher deposit spreads, corroborating our evidence

¹⁹In Table A.6, we also examine bank advertising spending using the same specification.

in Table 3 and Table 5. Service content redirects depositor attention from rates to quality attributes, reducing effective rate sensitivity and enabling wider equilibrium spreads.

Our second measure of deposit market power is deposit beta, which captures pass-through: the sensitivity of a bank’s deposit rate to changes in the Fed funds rate. Deposit beta is calculated by regressing changes in a bank’s deposit rate on contemporaneous and lagged changes in the Fed funds rate and summing the coefficients. A bank with high market power has low deposit beta—when the Fed raises rates, such a bank passes through little of the increase to depositors, preserving its spread. Conversely, a bank with weak market power, such as one reliant on rate-sensitive wholesale funding, has a beta close to 1 and earns minimal spread (Drechsler et al., 2021). Column (4) shows that banks emphasizing pricing in advertising have significantly higher deposit betas, indicating that their depositors’ rates are more sensitive to Fed funds rate changes. This pattern accords with our earlier evidence in Table 5 and reflects pricing-focused advertising attracting rate-sensitive, switching-prone depositors.

Column (5) uses Tobin’s q as a summary measure of franchise value (Demsetz et al., 1996). The insignificant coefficients indicate that while advertising content generates measurable deposit market power (captured by spreads and beta), these effects do not translate into sustained differences in long-run firm valuation. This may reflect that advertising-driven deposit advantages are contestable—competitors can adjust their own content strategies—or that Tobin’s q is a noisy valuation measure for diversified bank holding companies with many business lines beyond retail deposits.

To summarize, advertising content is associated with measurable differences in deposit market power—captured by spreads and deposit betas—consistent with advertising serving as a mechanism through which banks invest in building a deposit franchise. Together with the advertising spending results in Table A.6, this evidence demonstrates that both the content and intensity of advertising shape deposit market power—a key dimension of bank franchise value.

B. Transmission of Monetary Policy

Having established that advertising content drives variation in deposit market power, we now ask whether this heterogeneity alters how banks transmit monetary policy shocks to the real economy. If non-price advertising reduces depositors’ effective rate sensitivity, banks with strong advertising-built franchises should exhibit different deposit and lending responses to

Fed funds rate changes. We begin with aggregate time-series patterns. Figure 4 plots how banks’ advertising content evolves over monetary policy cycles, with NBER-dated recessions shaded.

[Insert Figure 4 Here.]

Figure 4 documents systematic cyclical patterns in advertising content. During tightening cycles, banks shift emphasis away from pricing and service messaging toward emotional appeals and trust-building—the persuasive channel. Conversely, during the zero-lower-bound period (2008Q4–2015Q3), banks prominently feature pricing and financial information. These patterns reflect banks’ dynamic response to the deposit competition environment: when rates rise, trust content may help retain depositors amid intensifying rate competition; when rates are compressed, pricing content becomes more prominent as banks compete for rate-sensitive depositors in a low-spread environment. While the aggregate time series does not establish causality, it documents the timing and magnitude of these shifts, motivating the within-county analysis that follows.

To examine how advertising content shapes monetary policy transmission, we implement a within-county specification following Drechsler et al. (2017) on the same bank holding company (b)-county (c)-year (t) panel as in Section IV.B:

$$Y_{b,c,t} = \beta^{\text{Ad}} \text{AdContent}_{b,c,t-1} + \gamma^{\text{Ad}} \Delta \text{FF}_t \times \text{AdContent}_{b,c,t-1} + \beta^{\text{HHI}} \text{Deposit-HHI}_{b,t-1} + \gamma^{\text{HHI}} \Delta \text{FF}_t \times \text{Deposit-HHI}_{b,t-1} + \eta_{b,c} + \eta_{c,t} + \epsilon_{b,c,t}, \quad (11)$$

where $Y_{b,c,t}$ includes the change in deposit spread, growth rate of deposits, and mortgage and small business loan origination for bank b at quarter t . $\text{AdContent}_{b,c,t-1}$ are lagged advertising content of bank b in county c in year $t-1$. ΔFF_t is the contemporaneous change in the Fed funds rate at t . $\text{Deposit-HHI}_{b,t-1}$ is the lagged bank-level deposit concentration for bank b in year $t-1$. $\eta_{b,c}$ and $\eta_{c,t}$ are bank-county and county-year fixed effects, respectively. We cluster standard errors at the county level.

The coefficient of interest is γ^{Ad} , the interaction between advertising content and Fed funds rate changes, which measures whether a bank’s balance-sheet response to a monetary policy shock depends on the composition of its advertising. To isolate advertising-driven heterogeneity from the deposit market concentration channel studied by Drechsler et al. (2017), we control for $\text{Deposit-HHI}_{b,t-1}$ and its interaction with ΔFF_t . County-year fixed effects ($\eta_{c,t}$) absorb time-varying local demand conditions, so we compare outcomes of different

banks operating in the same county and year but with different advertising content. Bank-county fixed effects ($\eta_{b,c}$) absorb time-invariant characteristics of each bank’s local presence, including historical market share, branch networks, and deposit base composition.

[Insert Table 7 Here.]

Table 7 shows that banks with different advertising content display heterogeneous balance-sheet responses to Fed funds rate increases. Banks emphasizing the informative (service) channel exhibit larger increases in deposit spreads when the Fed raises rates (column 1), reflecting their higher baseline market power and wider spread adjustment. Correspondingly, these service-focused banks experience larger deposit outflows (column 2) and greater declines in mortgage and small-business lending (columns 3–4). This pattern aligns with the bank-level evidence in Table 6 column (3): service-focused banks maintain higher deposit spreads and thus higher deposit market power, amplifying the balance-sheet contraction when rates rise.

These results accord with the bank deposit channel (Drechsler et al., 2021). In our framework, non-price advertising builds deposit franchise value by reducing depositors’ effective rate sensitivity—through attention redirection (service content) or emotional attachment (trust content). Banks with strong advertising-driven franchises sustain market power over retail deposits: when the Fed raises rates, their depositors’ rates adjust slowly, widening spreads and eventually triggering deposit outflows that constrain lending. Advertising content thus operates as a distinct source of bank heterogeneity in monetary policy transmission, complementing the market concentration channel documented by Drechsler et al. (2017).

C. Identifying the Impacts of Bank Advertising

The preceding analyses document associations between advertising content and deposit market power, but advertising is a bank’s strategic choice, raising concerns about endogeneity. Equilibrium forces may cause advertising to correlate with demand or performance for reasons other than the direct effect of advertising. Banks may also target advertising based on anticipated rather than realized product demand, creating reverse causality. We address these concerns using a border-discontinuity design that identifies the causal effect of advertising content on financial product demand.

Our empirical specification uses a border-discontinuity design, following Shapiro (2018) and Shapiro (2020). The design exploits geographic discontinuities at the borders of Dis-

nated Market Areas (DMAs), leveraging the discrete nature of local television markets. Consumers on opposite sides of a DMA border face different advertising exposure from the same bank due to the geography of the media market, while sharing similar demographics, income levels, and access to financial products. To verify this identifying assumption, Appendix Table A.10 reports balance tests showing that observable county characteristics—including average income, population, gender, race, age, and education—are not systematically associated with advertising content at the border.

[Insert Figure 5 Here.]

To illustrate, Figure 5 shows the Pittsburgh and Johnstown-Altoona DMAs in the state of Pennsylvania, with the border region highlighted in bold. The design compares how demand for the same bank’s financial products differs between counties on opposite sides of this border, where the bank operates branches on both sides but faces different DMA-specific television advertising exposure.

Consider a bank (b)-county (c)-border ($d(c)$)-year (t) panel. Let $d(c)$ stand for the border group to which county c belongs. We estimate the causal effect of advertising on the demand for financial products using the following specifications:

$$Y_{b,c,t} = \beta \cdot \text{AdContent}_{b,c,t} + \eta_{b,c} + \eta_{b,d(c),t} + \epsilon_{b,c,d(c),t}, \quad (12)$$

where $Y_{b,c,t}$ is a set of outcome variables for various financial products, such as the growth rate of deposits, change in deposit spread, and mortgage and small business loan origination, of bank b in county c at t . The independent variables of interest are the advertising content shares: pricing and service quality, with emotion/trust as the omitted baseline category.

The fixed effects are chosen to isolate advertising content variation from potential confounders. Bank-county fixed effects $\eta_{b,c}$ absorb all time-invariant characteristics of a bank’s local presence—branch density, historical market share, deposit base composition—preventing cross-county differences in bank fundamentals from biasing the estimates. Bank-border-year fixed effects $\eta_{b,d(c),t}$ absorb bank-specific and border-specific demand trends in each year, ensuring we do not conflate time-varying bank-wide strategy shifts with local advertising effects. With both sets of fixed effects, we identify the intensive-margin effect of advertising content by comparing the same bank’s product demand across counties within a border group that have differential advertising exposure. The identifying assumption is that, conditional

on these fixed effects, the same bank would experience parallel trends in financial product demand on both sides of a DMA border absent differential advertising.

The analysis is restricted to bank-county observations with positive advertising spending, so the estimated coefficients capture the intensive-margin effect of advertising content conditional on the bank choosing to advertise in that market. Because counties near DMA boundaries may belong to multiple border groups, the same bank-county-year observation can appear multiple times with different border identities. To account for correlation within bank-county pairs and within border-year groups, we cluster standard errors two-way at the bank-county and border-year levels.

[Insert Table 8 Here.]

Table 8 presents the results. Columns (1) and (2) examine how advertising content causally affects deposit demand and pricing. Banks emphasizing the informative (service) channel experience 1.3 percentage-point higher deposit growth—a sizable effect representing 34% of the sample mean quarterly deposit growth of 3.8%. Service-focused advertising also enables significantly higher deposit spreads, with a coefficient of 1.9 percentage points. Given that the sample mean deposit spread is -0.72 percentage points—negative because Fed funds rates were near zero during much of the sample period—this effect is economically large. In contrast, banks emphasizing pricing in their advertising experience lower deposit growth and narrower spreads, consistent with pricing content attracting rate-sensitive depositors and eroding franchise value. These causal estimates reinforce the cross-sectional patterns documented in the preceding sections: the informative (service) channel builds deposit franchise value by redirecting depositor attention from rates to quality.

Columns (3) and (4) reveal a striking deposit–loan asymmetry. While informative (service) content drives deposit market power, it is the persuasive (trust) channel—emotion and trust-building, the omitted baseline—that drives mortgage and small-business loan origination. Banks emphasizing trust attract significantly higher loan demand. This deposit–loan asymmetry aligns with our framework’s asset-side logic: borrowers cannot assess loan quality *ex ante*, so the informative channel is weaker, and trust content plays a larger role in reducing effective rate sensitivity and building lending relationships. Appendix Table A.8 shows that advertising *spending* also drives deposit and loan volumes, demonstrating that both content and intensity matter for financial product demand.

VI. Conclusion

This paper shows that banks compete through advertising content—pricing, service (informative), and trust (persuasive)—and that this content is a mechanism for building deposit franchise value and shaping monetary policy transmission. We develop a framework in which advertising content determines depositors’ rate sensitivity through sorting. Empirically, high-market-share banks prioritize non-price content and adjust it systematically in response to entry, mergers, FinTech competition, and trust shocks. A DMA border-discontinuity design identifies causal effects: service content drives deposit growth and wider spreads, while trust content drives loan demand—reflecting our prediction that trust matters more on the asset side, where loan quality is less transparent.

Advertising content also operates as a source of heterogeneity in monetary policy transmission, complementing the market concentration channel of [Drechsler et al. \(2017\)](#). Banks emphasizing non-price content attract less rate-sensitive depositors and exhibit lower deposit betas, consistent with our framework. Unlike prior work that takes depositor composition as exogenous, we show that banks shape it through content choice—a mechanism through which some banks accumulate “sleepy” depositor bases ([Egan et al., 2025](#)). The border design establishes causality for deposit growth and spreads; the deposit beta channel is guided by our model and consistent with observed patterns.

Open questions remain. Our framework is deliberately parsimonious (two banks, static content) and omits dynamic adjustment and general equilibrium feedback. Welfare implications are ambiguous: service content may improve match quality while trust content may exploit behavioral biases ([Anderson and Renault, 2006](#)). As [Bertrand et al. \(2010\)](#) emphasize, theory offers limited guidance on which content dimensions matter; our taxonomy is a step forward, but deeper foundations are needed, especially to explain why service matters for deposits while trust matters for loans.

Methodologically, we show that unsupervised video embedding clustering robustly encodes multimodal advertising content ([Gentzkow et al., 2019](#)). As banking competition migrates to digital channels, this framework scales to richer content data—enabling new research on how financial institutions build and defend customer franchises.

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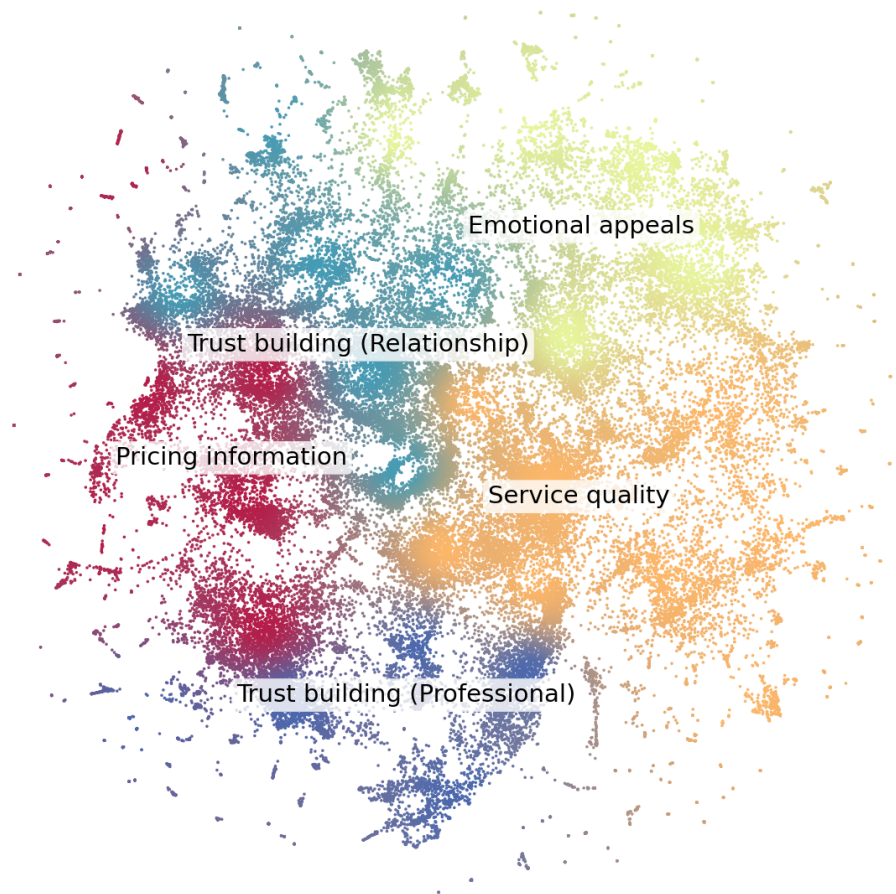
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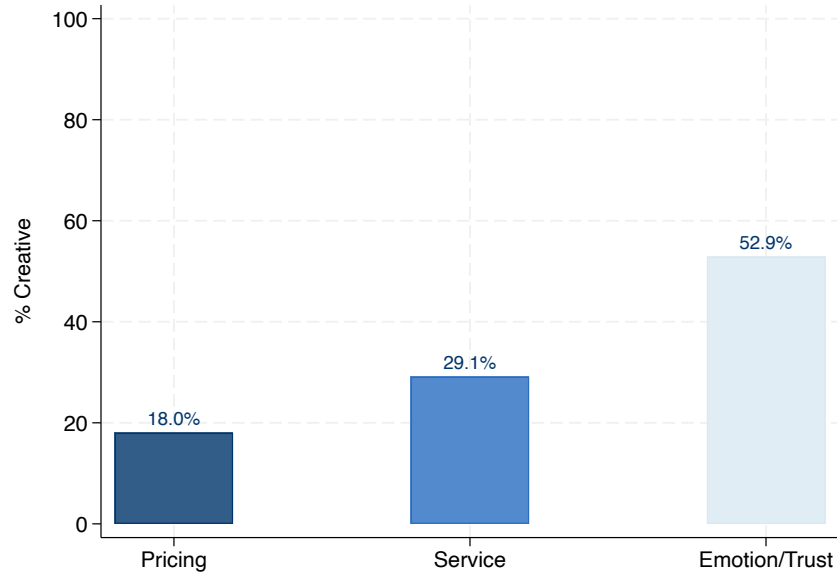
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Figure 1. Visualization of Video Embeddings and Advertising Content Clusters



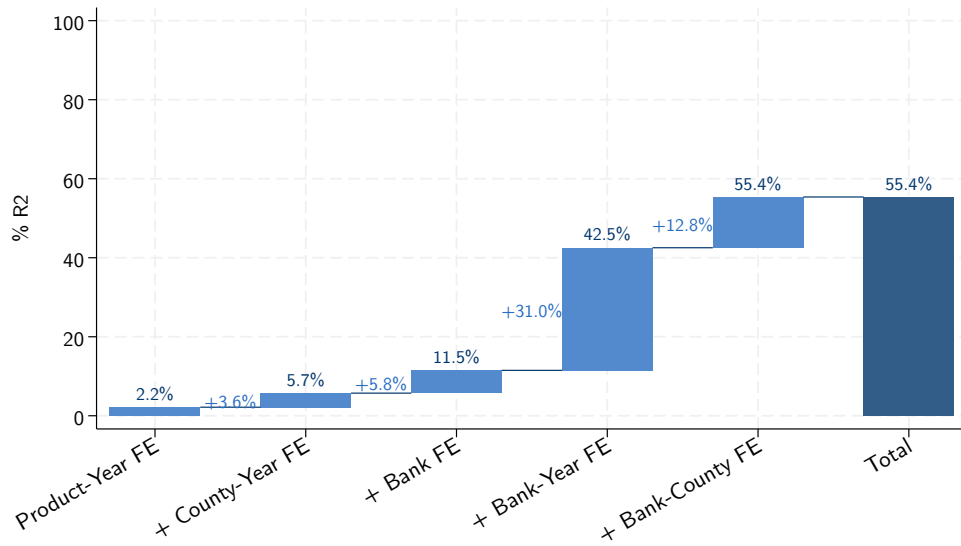
Notes. This figure shows the visualization of the two-dimensional embeddings of bank advertising videos, as described in Section A. It shows three categories of advertising content: financial information, service quality, and trust-building and life aspirations (including relationship building, professional imagery, and emotional appeal).

Figure 2. Distribution of Bank Advertising Content



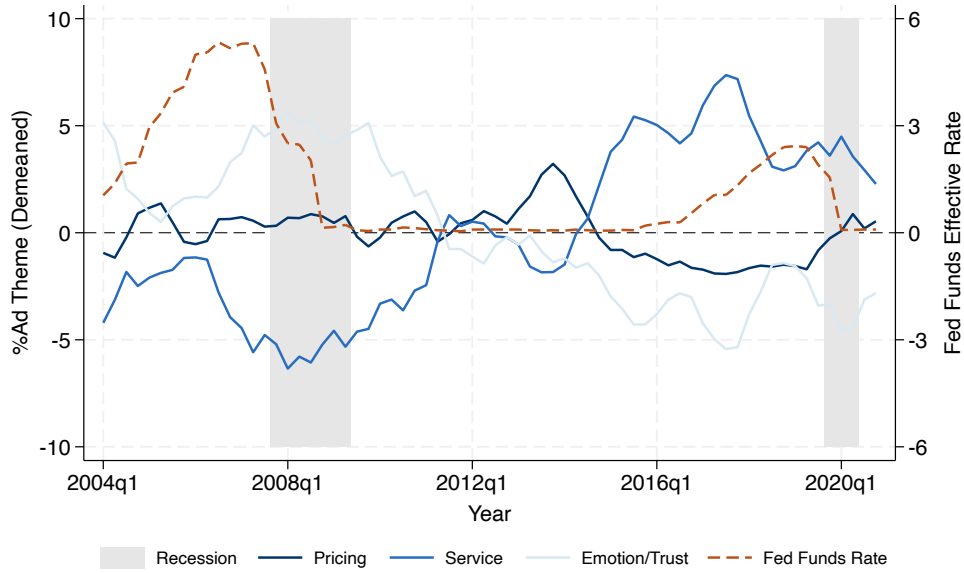
Notes. This figure shows the distribution of bank advertising content, including financial information, service quality, and emotional appeal/trust-building.

Figure 3. Decomposing Variations of Advertising Content



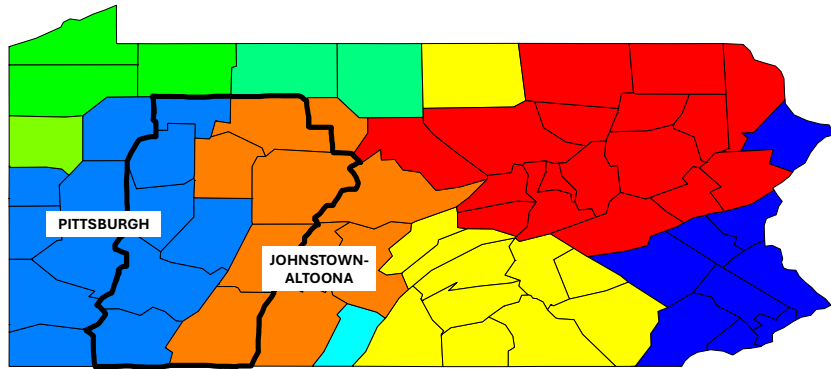
Notes. This figure shows the percentages of variations for advertising content can be explained by different specifications of fixed effects. We implement a Shapley-Owen decomposition on a bank-county-product-year panel. We consider the following set of fixed effects: product-year fixed effects, county-year fixed effects, bank fixed effects, bank-year fixed effects, and bank-county fixed effects.

Figure 4. Advertising Content and Monetary Policy



Notes. This figure plots the distribution of bank advertising content (e.g., financial information, service quality, and trust-building) alongside the Fed Funds rate over time. The NBER dated-recessions are shaded. The advertising content variables are demeaned by the sample average.

Figure 5. Example of DMA Border Region



Notes. This figure shows the DMA map for Pennsylvania, highlighting the border region between Pittsburgh and Johnstown-Altoona used in border discontinuity design, as in Equation (12).

Table 1. Summary Statistics

	count	mean	sd	min	p25	p50	p75	max
<i>Panel (a): Creative Level</i>								
Duration	51,349	27.40	13.88	1	15	30	30	141
Air Months	51,349	13.86	25.92	0	1	4	14	216
# Schedule Times, Local TV	49,262	1,559.96	14,311.59	1	28	94	371	2,114,381
# Schedule Times, National TV	10,525	1,340.63	4,802.44	1	30	256	1,118	197,505
Category: Personal Banking	50,910	24.16	41.84	0.00	0.00	0.00	22.66	100.00
Category: Business Banking	50,910	30.53	45.57	0.00	0.00	0.00	99.97	100.00
Category: Personal Credit	50,910	8.00	26.65	0.00	0.00	0.00	0.00	100.00
Category: Mortgage	50,910	17.08	37.12	0.00	0.00	0.00	0.00	100.00
Category: Investment/Retirement	50,910	20.23	39.88	0.00	0.00	0.00	0.00	100.00
<i>Panel (b): Bank-Year Level</i>								
Deposits (MM)	6,569	15,048.44	92,022.53	0.00	270.93	692.91	2,546.90	1,718,968.13
Mortgage Amount (MM)	5,998	2,180.74	14,202.02	0.00	26.44	96.87	396.72	332,987.09
Business Loan Amount (MM)	2,984	911.14	2,865.80	0.00	69.13	154.99	448.98	40,389.83
CFPB Complaints per MM Deposits	6,568	0.02	0.11	0.00	0.00	0.00	0.00	5.48
Saving APY (%)	4,807	0.62	0.74	0.01	0.14	0.29	0.89	5.04
Time APY (%)	4,903	1.36	1.36	0.03	0.31	0.66	2.26	5.36
Deposit HHI	6,568	0.21	0.10	0.05	0.14	0.19	0.24	0.96
Deposit Beta	5,924	0.40	0.11	0.02	0.34	0.39	0.44	1.45
Deposit Productivity	1,287	0.49	1.03	-1.69	-0.29	0.36	1.24	2.90
Asset Productivity	1,287	0.26	1.05	-11.72	-0.19	0.26	0.74	4.21
Tobin's q	1,739	1.18	0.37	0.40	0.97	1.12	1.32	6.59
<i>Panel (c): Bank-County-Year Level</i>								
Deposits (MM)	94,238	909.85	7,728.93	0.00	43.95	105.16	323.01	626,247.20
Deposits Share (%)	94,229	31.98	30.14	0.00	8.57	22.10	45.30	100.00
Saving Spread (%)	62,763	0.70	1.26	-2.09	0.01	0.10	0.95	4.92
Time Spread (%)	64,651	-0.21	0.83	-3.62	-0.43	0.01	0.20	1.20
Mortgage Amount (MM)	515,639	19.21	163.98	0.00	0.13	0.74	4.62	25,389.18
Mortgage Amount Share (%)	515,052	7.36	11.82	0.00	0.34	2.69	9.08	100.00
Business Loan Amount (MM)	468,278	4.53	27.39	0.00	0.06	0.27	1.38	3,040.95
Business Loan Amount Share (%)	468,265	9.65	14.05	0.00	1.32	4.22	11.55	100.00

Notes. This table presents summary statistics for creative-level data in Panel (a), bank-year level data in Panel (b), and bank-county-year level data in Panel (c). Panel (a) presents data on bank advertising derived from Nielsen's Ad Intel dataset. Duration is the length of an advertising in seconds. Air Months is the total number of months the advertising aired on television. # Schedule Times, Local TV / National TV is the total number of times an advertising was scheduled to air on local or national television. The five categories, including Personal Banking, Business Banking, Personal Credit, Mortgage, Investment/Retirement, represent the percentage distribution of the product categories in the advertising videos. Panel (b) contains a bank holding company-year panel combining various sources of financial intermediaries data, with unavailable data for any bank-year observation recorded as missing. Deposits is the amount of deposits from FDIC Summary of Deposits. Mortgage Amount is the total amount of mortgage loans originated from HMDA. Business Loan Amount is the total amount of business loans issued from CRA. CFPB Complaints per MM deposits is the number of complaints received from CFPB Consumer Complaint Database divided by the amount of deposits. Saving APY and Time APY is the average APY for savings and time deposit account, respectively. Deposit HHI is the bank-level market concentration, calculated as the deposit-weighted average of the county-level Herfindahl-Hirschman Index across all the counties that the bank holding company operates in. Deposit Beta is obtained from Drechsler et al. (2017) and Drechsler et al. (2021). Deposit Productivity and Asset Productivity are obtained from Egan et al. (2022). Tobin's q is the market value of assets divided by the book value of assets, computed from Compustat. Panel (c) contains a bank holding company-county-year panel. Deposits Share, Mortgage Amount Share, and Business Loan Amount Share are banks' market share of deposits, mortgage loans, and business loans in the county, respectively. Saving Spread and Time Spread are the difference between the Fed Funds Rate and the average APY for savings and time deposit accounts, respectively.

Table 2. Categories of Bank Advertising Content

Broad	Intermediate	Detailed	Related Keywords
Financial information	Interest rate	Rate & payment Loan rate	rate, low, fixed, APR, payment, monthly payment rate, loan, refinance, mortgage rate, lowest
	Payment plan	Monthly payment Loan payment	deposit, pay, monthly payment, fixed, low free, loan, credit, payment, income
	Cashback rewards	Account offer Cashback rewards	free, account, rate, APY, earn, minimum balance earning, cash, refund, checking account
Service quality	Local service	Local service Customer service Branch service	local, community, relationship, client, team customer, community bank, service, banker branch, building, location, state bank, ATM
	Online service	Mobile banking Conversational service	mobile, mobile banking, online, easier conversation, talk, customer, explain, representative
	Daily convenience	Credit report Cash transfer Everyday transactions	credit, score, free credit, checking credit app, send money, pay, mobile, wallet, easy card, ATM, store, debit card, purchase, traveling
Emotional appeal and trust building	Emotional appeals	Small business support	small business, build, grow, dream, success
		Home & future	life, home, family, moment, help, future
		College & education	college, child, education, school, plan, kid, student
		Game & sport sponsorship	game, player, team, fan, golf, basketball
		Sport sponsorship	player, sport, football, golf, basketball
	Relationship building	Investment & stock market	trade, stock, ETF, risk, fund investment, investing
		Brand promotion	trust, bank logo, member FDIC, national bank, news
		Home & community trust	home, community, dream, life, help, car, people
		Credit improvement	credit score, bad credit, financial, reward, cashback
		Family trust	family, picture, baby, kid, trust, Christmas
	Professional imagery	Patriotism trust	veteran, new day, USA, military, hero, American
		Advisory trust	firm, financial, advice, trust, planning, guidance
		Professional expertise	tax, income, money, market, strategy, social security
		Commodity market	gold, silver, capital, coin, metal, investment
		Retirement financial	senior, older, retirement, tax free, homeowner
		Wealth management	financial, planning, client, wealth, advisor, goal

Notes. This table shows the categories of bank advertising content and their associated keywords obtained from unsupervised clustering of video embeddings. The classification hierarchy has three levels: broad, intermediate, and detailed. The last column (Related Keywords) reports representative words with the highest average TF-IDF scores for each detailed category. Refer to Section A for more details.

Table 3. Advertising Content by Local Market Power

Ad Category	(1)	(2)	(3)	(4)	(5)	(6)
	Deposit			Mortgage		
% AdContent _{b,c,t}	Pricing	Service	Emotion/Trust	Pricing	Service	Emotion/Trust
<i>Panel (a): Local Market Share as Measure of Market Power</i>						
Local Market Share _{b,c,t-1}	-0.355** (0.141)	3.058*** (0.218)	-2.703*** (0.175)	-3.678*** (0.135)	-2.326*** (0.230)	6.003*** (0.213)
R-squared	0.053	0.068	0.067	0.063	0.067	0.068
<i>Panel (b): Market Share/Top Share as Measure of Market Power</i>						
Market Share/Top Share _{b,c,t-1}	-0.351*** (0.097)	2.674*** (0.157)	-2.323*** (0.127)	-1.922*** (0.057)	-0.568*** (0.102)	2.490*** (0.096)
R-squared	0.053	0.068	0.068	0.064	0.067	0.067
<i>Panel (c): Bank HHI as Measure of Market Power</i>						
Bank HHI _{b,c,t-1}	-5.354*** (0.060)	7.512*** (0.104)	-2.157*** (0.098)	-2.736*** (0.524)	-8.401*** (0.849)	11.137*** (0.689)
R-squared	0.061	0.072	0.067	0.062	0.067	0.067
Observations	922,811	922,811	922,811	706,796	706,796	706,796
No. of Institutions	937	937	937	721	721	721
Mean of Y	13.44	36.16	50.40	14.51	34.60	50.89
County-Year FE	Y	Y	Y	Y	Y	Y

Notes. This table examines the advertising content across banks with different levels of local market power in a bank(*b*)-county(*c*)-year(*t*) panel. The analysis considers two different products, including deposits in Columns (1)-(3), and mortgages in Columns (4)-(6). % Ad Content_{b,c,t} is the share of advertising content for that specific product used by bank *b* in county *c* in year *t*. The analysis considers three measures of local market power in the county *c* in the respective product market in year *t* - 1, including local market share in Panel (a), the ratio of local market share divided by the maximum market share in Panel (b), and bank-level HHI calculated as the product-weighted average of the county-level HHI across all the counties in year *t* - 1 in Panel (c). The model includes county-year fixed effects. Standard errors are clustered at the county level. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 4. Advertising Content After Economic Shocks

	(1)	(2)	(3)	(4)	(5)	(6)
Ad Category	All			All		
% AdContent _{b,c,t}	Pricing	Service	Emotion/Trust	Pricing	Service	Emotion/Trust
<i>Panel (a): Bank Entry Into Local Markets</i>						
Post(Branch Entry) _{b,c,t}	0.318*** (0.079)	-0.801*** (0.138)	0.484*** (0.117)			
Post(M&A Entry) _{b,c,t}				-6.799*** (0.538)	7.833*** (1.002)	-1.034 (0.767)
Observations	2,786,104	2,786,104	2,786,104	2,786,104	2,786,104	2,786,104
R-squared	0.209	0.226	0.204	0.209	0.226	0.204
No. of Institutions	2,401	2,401	2,401	2,401	2,401	2,401
Mean of Y	13.97	36.61	49.41	13.97	36.61	49.41
Bank FE	Y	Y	Y	Y	Y	Y
County-Year FE	Y	Y	Y	Y	Y	Y
<i>Panel (b): Incumbent Responses to FinTech Entry</i>						
Post(FinTech Entry) _{c,t}	-0.544*** (0.052)	0.615*** (0.094)	-0.071 (0.079)			
FinTech Market Share _{c,t}				-0.382** (0.176)	0.981*** (0.310)	-0.598** (0.281)
Observations	2,786,104	2,786,104	2,786,104	2,786,104	2,786,104	2,786,104
R-squared	0.477	0.480	0.449	0.477	0.480	0.449
No. of Institutions	2,401	2,401	2,401	2,401	2,401	2,401
Mean of Y	13.97	36.61	49.41	13.97	36.61	49.41
County FE	Y	Y	Y	Y	Y	Y
Bank-Year FE	Y	Y	Y	Y	Y	Y
<i>Panel (c): Wells Fargo Scandal</i>						
Post(Scandal) _t	-2.781*** (0.102)	-4.915*** (0.159)	7.697*** (0.188)			
Post(Scandal) _t × 1(Wells Fargo Presence) _c				-0.079*** (0.027)	-0.055 (0.047)	0.133*** (0.044)
Sample	Wells Fargo	Wells Fargo	Wells Fargo	Other Banks	Other Banks	Other Banks
Observations	55,612	55,612	55,612	2,730,492	2,730,492	2,730,492
R-squared	0.067	0.046	0.059	0.483	0.485	0.452
No. of Institutions	1	1	1	2,400	2,400	2,400
Mean of Y	12.77	33.86	53.37	14	36.67	49.33
County FE	Y	Y	Y	Y	Y	Y
Bank-Year FE	N	N	N	Y	Y	Y

Notes. This table examines how bank advertising content changes after bank and FinTech entry into local markets and a bank scandal, using a bank(*b*)-county(*c*)-year(*t*) panel. Panel (a) reports effects of bank's entry into local markets, through opening a new branch in Columns (1)-(3) or M&A in Columns (4)-(6). Post(Branch Entry)_{b,c,t} and Post(M&A Entry)_{b,c,t} are dummy variables for whether year *t* falls within the three-year window post bank *b*'s entry into county *c* through opening a new branch and through M&A, respectively. Panel (b) reports effects of FinTech's entry into local markets. Post(FinTech Entry)_{c,t} is a dummy variable for whether the year *t* falls within the three-year window post the FinTech's entry into county *c*. FinTech Market Share_{c,t} is the mortgage market share of FinTech companies in county *c* and year *t*. Panel (c) reports the effects of the Wells Fargo scandal in 2016. Columns (1)–(3) focus on Wells Fargo itself, and Columns (4)–(6) examine all other banks excluding Wells Fargo. Post(Scandal)_t is a dummy variable for whether year *t* falls in the range of 2016 to 2020. 1(Wells Fargo Presence)_c is a dummy variable for whether county *c* has at least one Wells Fargo branch in 2016. Fixed effects are denoted at the bottom. Standard errors are clustered at the county level. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 5. Advertising Content by High- and Low-Rate and Service Quality Banks

	(1)	(2)	(3)	(4)	(5)	(6)
Ad Category	Deposit			Deposit		
% AdContent _{b,c,t}	Pricing	Service	Emotion/Trust	Pricing	Service	Emotion/Trust
High Deposit Rate _{b,c,t-1}	3.129*** (0.039)	-2.219*** (0.070)	-0.910*** (0.054)			
High Service Quality _{b,c,t-1}				-0.488*** (0.081)	2.448*** (0.169)	-1.960*** (0.141)
Observations	665,019	665,019	665,019	987,169	987,169	987,169
R-squared	0.078	0.073	0.078	0.053	0.059	0.060
No. of Institutions	922	922	922	1,120	1,120	1,120
Mean of Y	13.63	35.98	50.38	13.30	36.41	50.28
County-Year FE	Y	Y	Y	Y	Y	Y

Notes. This table examines the advertising content across banks with different levels of deposit rate and service quality in a bank(*b*)-county(*c*)-year(*t*) panel. The analysis considers the deposit business. % AdContent_{b,c,t} is the share of advertising content of bank *b* in county *c* in year *t*. In Columns (1)-(3), High Deposit Rate_{b,c,t-1} is a dummy variable for whether bank *b* is above the 80 percent quantiles of deposit rate rankings among all the banks in the county *c* in year *t* - 1. We rank banks based on the interest rate of time deposit (i.e., \$10,000 12-month CDs), saving deposit (i.e., \$25,000 money market deposit accounts), and checking deposit (i.e., interest checking accounts with less than \$2,500) in each year, and then average these ranks as the overall deposit rate ranking for the bank in a given year. In Columns (4)-(6), High Service Quality_{b,c,t-1} is a dummy variable for whether bank *b* is above the 80 percent quantiles of service quality among all the banks in the county *c* in year *t* - 1. We measure service quality using number of CFPB consumer complaints scaled by deposits for the banks in a given year. The model includes county-year fixed effects. Standard errors are clustered at the county level. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 6. Advertising Content and Bank Value

	(1)	(2)	(3)	(4)	(5)
Ad Category	Deposit	Asset	Deposit	Deposit	All
<i>Y</i> _{b,t}	Deposit TFP	Asset TFP	Deposit Spread	Deposit Beta	Tobin's <i>q</i>
1(Pricing) _{b,t-1}	0.389** (0.181)	-0.903*** (0.266)	0.035 (0.098)	0.018** (0.009)	0.041 (0.056)
1(Service) _{b,t-1}	-0.168 (0.108)	-0.445*** (0.163)	0.101* (0.058)	0.005 (0.005)	-0.012 (0.023)
1(Emotion/Trust) _{b,t-1}			<i>Omitted</i>		
Observations	2,638	2,638	8,053	9,348	6,239
R-squared	0.930	0.214	0.509	0.148	0.345
No. of Institutions	165	165	537	810	325
Mean of Y	0.793	0.528	0.169	0.408	1.157
Bank Controls	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y

Notes. This table examines the relationship between bank advertising content and bank value in a bank(*b*)-quarter(*t*) panel. The sample contains all the publicly listed U.S. bank holding companies with non-missing and positive Advertising and Marketing Expenses (Compustat item **xadq**). The outcome variables are bank deposit productivity (Deposit TFP) and asset productivity (Asset TFP) in Columns (1)-(2) from Egan et al. (2022), deposit spread and deposit beta in Columns (3)-(4) from Drechsler et al. (2021), and Tobin's *q*. The measures of advertising content include pricing information 1(Pricing)_{b,t-1} and service quality 1(Service)_{b,t-1}, omitting emotional appeal and trust building 1(Emotion/Trust)_{b,t-1}. The model includes quarter fixed effects and lagged bank controls. Lagged bank controls include the log of the bank's total assets, the bank's Z-Score (the bank's ROA plus its equity ratio divided by its standard deviation of ROA), the bank's ROA, the bank's equity to assets ratio, the bank's deposits to assets ratio, and the bank's deposit and mortgage market concentration HHI, following Gelman et al. (2023) and Egan et al. (2022). Standard errors are clustered at the bank level with a wild cluster bootstrap procedure (1000 times). *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 7. Advertising Content and Monetary Policy Transmission

	(1)	(2)	(3)	(4)
$Y_{b,c,t}$	$\Delta(\text{Deposit Spread})$	$\Delta \ln(\text{Deposits})$	$\ln(\text{Mortgage})$	$\ln(\text{Small Business Loan})$
1(Pricing) $_{b,c,t-1}$	-0.0544*** (0.0128)	0.0020 (0.0039)	-0.1071** (0.0479)	0.2939*** (0.0182)
1(Service) $_{b,c,t-1}$	0.0411*** (0.0060)	0.0078*** (0.0025)	-0.3597*** (0.0293)	0.1268*** (0.0101)
$\Delta FF_t \times 1(\text{Pricing})_{b,c,t-1}$	0.0753*** (0.0250)	0.0078** (0.0038)	-0.2371*** (0.0449)	-0.1610*** (0.0193)
$\Delta FF_t \times 1(\text{Service})_{b,c,t-1}$	0.0991*** (0.0106)	-0.0088*** (0.0023)	-0.3645*** (0.0241)	-0.0660*** (0.0102)
1(Emotion/Trust) $_{b,c,t-1}$ and $\Delta FF_t \times 1(\text{Emotion/Trust})_{b,c,t-1}$	<i>Omitted</i>			
Observations	51,516	86,829	684,850	485,658
R-squared	0.915	0.492	0.643	0.830
No. of Institutions	603	809	1,019	351
Mean of Y	0.140	0.0567	11.89	12.48
HHI $_{b,t-1}$ and $\Delta FF_t \times \text{HHI}_{b,t-1}$	Y	Y	Y	Y
County-Year FE	Y	Y	Y	Y
Bank-County FE	Y	Y	Y	Y

Notes. This table examines the relationship between advertising content and the transmission of monetary policy using a bank(b)-county(c)-year(t) panel. The lagged measures of advertising content for bank b in county c in year $t - 1$ are pricing information 1(Pricing) $_{b,c,t-1}$ and service quality 1(Service) $_{b,c,t-1}$, omitting emotional appeal and trust building 1(Emotion/Trust) $_{b,c,t-1}$. ΔFF_t is the contemporaneous change in the Fed funds rate. Following Drechsler et al. (2017), we control for bank-level Deposit-HHI $_{b,t-1}$ and its interaction term $\Delta FF_t \times \text{HHI}_{b,t-1}$. Bank-level Deposit-HHI $_{b,t-1}$ is calculated as the deposit-weighted average of the county-level HHI across all the counties where the bank b operates in year $t - 1$. The model includes county-year fixed effects and bank-county fixed effects. Standard errors are clustered at the county level. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 8. Advertising Content and Financial Product Demand:
Border Discontinuity Design

	(1)	(2)	(3)	(4)
Ad Category	Deposit		Mortgage	Loan
$Y_{b,c,t}$	$\Delta \ln(\text{Deposits})$	$\Delta(\text{Deposit Spread})$	$\ln(\text{Mortgage})$	$\ln(\text{Small Business Loan})$
1(Pricing) $_{b,c,t}$	-0.019 (0.014)	-0.171*** (0.022)	-0.847*** (0.099)	-0.067** (0.028)
1(Service) $_{b,c,t}$	0.0131* (0.008)	0.0194** (0.009)	-0.810*** (0.060)	-0.018 (0.017)
1(Emotion/Trust) $_{b,c,t}$	<i>Omitted</i>			
Observations	48,709	48,709	335,171	401,296
R-squared	0.356	0.954	0.681	0.784
No. of Institutions	352	352	930	418
Mean of Y	0.038	-0.007	11.64	12.35
Bank-County FE	Y	Y	Y	Y
Bank-Border-Year FE	Y	Y	Y	Y

Notes. This table examines the relationship between advertising content and demand for financial products using a border discontinuity design in a bank(b)-county(c)-border($d(c)$)-year(t) panel. The outcome variables, $Y_{b,c,t}$, include the changes in the log of deposits $\Delta \ln(\text{Deposits})$ in Column (1), changes in the deposit spread $\Delta(\text{Deposit Spread})$ in Column (2), the log amount of new mortgage origination $\ln(\text{Mortgage})$ in Column (3), and the log amount of new small business loan origination $\ln(\text{Small Business Loan})$ from CRA in Column (4). The measures of advertising content include pricing information 1(Pricing) $_{b,c,t}$ and service quality 1(Service) $_{b,c,t}$, omitting emotional appeal and trust building 1(Emotion/Trust) $_{b,c,t}$. We use product-specific advertising measures, namely deposits in Column (1)-(2), mortgages in Column (3), and business loans in Column (4). The model includes bank-county fixed effects and bank-border-year fixed effects. Standard errors are clustered at the bank-county level and border-year level. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Internet Appendix (For Online Publication Only)

How Do Banks Compete?

Evidence from Advertising Videos

Xugan Chen

Allen Hu

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A.1. Additional Results

A.1.1. Additional Sample Descriptions

This appendix provides additional descriptions of our sample of financial advertisements.

Figure A.1 shows that bank commercials have relatively short life cycles, and banks use different commercials in different DMAs. Panel (a) of Figure A.1 shows the distribution of air months for each creative video. 48% of creative videos were only shown on TV for less than three months. The median and average air months of financial commercials is four and fourteen months, respectively. This is robust for the sample of standard thirty-second commercials, both banks and non-banks, and different types of banks (such as those with different sizes, deposit rates, and HHIs). Panel (b) of Figure A.1 illustrates the geographic coverage of bank advertising. The dark blue line indicates the annual average number of DMAs where banks run advertising. The light blue line shows the average number of DMAs each commercial covers. The dark blue line is persistently higher than the light blue line. Such a pattern highlights the fact that banks run different commercials in different regions. Taken together, Figure A.1 documents the stylized facts that banks constantly update the content of their TV commercials, with the median life cycle of a creative being four months; banks actively use different commercials in different markets.

Table A.1 investigates which banks select into running advertising campaigns, and thus enter our sample of analysis. We use a bank(b)-quarter(t) sample of all the publicly listed U.S. bank holding companies at the quarterly frequency. The selection variable $1(\text{Advertising})_{b,t}$ is defined as whether bank b has non-missing and positive Compustat item Advertising and Marketing Expenses (`xadq`) in quarter t . We then use a comprehensive set of lagged bank characteristics to explain banks' selection into advertising, controlling for quarter fixed effects η_t . Therefore, our design exploits cross-sectional variations across banks to explain the sample selection. The explanatory variables are lagged one period and include log total assets, Z-Score (the bank's ROA plus its equity ratio divided by its standard deviation of ROA), ROA, Tobin's q , deposit spread, deposit and asset productivity from Egan et al. (2022), and the bank's deposit and mortgage market concentration HHI, calculated as the volume-weighted average of the county-level HHI across all the counties where the bank

operates.

The results indicate that there is no single variable that can consistently explain banks' decision to advertise, except for the deposit market HHI, whose coefficient is significantly negative. One potential interpretation is that when a bank faces more intense competition in the deposit market, as indicated by low market concentration and HHI, it chooses to run advertisements to attract customers and capture market share. There is some weaker evidence that larger banks are more likely to use advertising campaigns. ROA and Tobin's q seem to have negative correlations with the selection, suggesting that slightly weaker banks are more likely to use TV advertising campaigns, although the statistical significance is fragile. Deposit spread has a negative correlation with the selection, suggesting that banks with lower market power are more active in TV advertising.

A.1.2. Additional Results on Advertising Content

Figure A.2 and Figure A.3 provide detailed cluster visualization and distributions of advertising content. Banks may use different content for different products because they have different target customers and different marketing strategies. Figure A.4 shows the advertising content distribution across product categories. Table A.2 studies the transition probability of advertising content from one year to the next. The high probability of staying within the same content category (as indicated by the large diagonal terms in the transition matrix) suggests that banks often adhere to a fixed advertising content strategy in consecutive years.

A.1.3. Additional Results on Advertising Spending

Table A.6 examines the relationship between bank advertising spending and bank market power and franchise value. We use the bank holding company (b)-quarter (t) panel and estimate the following specifications:

$$Y_{b,t} = \beta \cdot \text{AdSpend}_{b,t-1} + \gamma X_{b,t-1} + \eta_t + \epsilon_{b,t}, \quad (\text{A1})$$

where $Y_{b,t}$ is a set of bank value measures such as productivity, deposit spread, and Tobin's q of bank b in quarter t . $\text{AdSpend}_{b,t-1}$ is the advertising expense from Compustat scaled by

total non-interest expense from Call Reports in quarter $t - 1$. $X_{b,t-1}$ is a list of lagged bank controls, including the log of the bank’s total assets, the bank’s Z-Score (the bank’s ROA plus its equity ratio divided by its standard deviation of ROA), the bank’s ROA, the bank’s equity to assets ratio, the bank’s deposits to assets ratio and the bank’s deposit and mortgage market concentration HHI, following [Gelman et al. \(2023\)](#) and [Egan et al. \(2022\)](#); η_t are time fixed effects. We use a sample with non-missing and positive advertising expenses. Standard errors are clustered at the bank level using wild bootstrap. This specification leverages the cross-sectional variation of banks, as some measures, such as bank productivity, are inherently cross-sectional in nature. To mitigate the reverse causality interpretation, we lag all advertising measures for one period, relative to the dependent variables.

Table [A.6](#) displays the results on advertising spending. Columns (1) and (2) show that banks with higher advertising expenses tend to have higher deposit and asset productivity, as measured in [Egan et al. \(2022\)](#). This finding is consistent with both customer-based and technology-based explanations of dispersion in bank productivity. Banks actively use advertising to strengthen their market power, target customers with diverse demographics, establish trust, and differentiate themselves from competitors.

Columns (3) to (5) indicate that financial advertising is positively correlated with the bank deposit power and franchise value. Column (3) documents a significant positive relationship between advertising and deposit spread, namely, banks with higher advertising expenses tend to have higher deposit spreads and, therefore, greater deposit market power. In terms of magnitude, a one percentage point increase in advertising expense is associated with a 7 basis point increase in deposit spread, whose mean value is 45 basis points. Column (4) shows an insignificant relationship between advertising and deposit beta. This pattern suggests that the variation in bank deposit market power can stem from multiple sources, including branches, salaries, and technology, and cannot be fully attributed to advertising alone. In column (5), we follow [Demsetz et al. \(1996\)](#) to use banks’ Tobin’s q as a proxy for banks’ franchise value. We find a significant positive correlation between banks’ Tobin’s q and advertising spending. Consistent with findings on banks’ deposit power in the market, efforts to craft images through advertising positively contribute to banks’ franchise value.

To summarize, the empirical evidence suggests that financial advertising contributes to

banks' franchise value as banks with higher advertising expenses tend to have higher deposit and asset productivity, higher deposit market power, and higher franchise value. Such findings suggest that advertising is an important way through which banks invest in building a deposit franchise, which gives them market power.

Table A.7 follows Drechsler et al. (2017) and shows how advertising spending affects monetary policy transmission. We estimate the following within-county specification on the same bank holding company (b)-county (c)-year (t) panel.

$$Y_{b,c,t} = \beta^{\text{Ad}} \text{AdSpend}_{b,c,t-1} + \gamma^{\text{Ad}} \Delta \text{FF}_t \times \text{AdSpend}_{b,c,t-1} + \beta^{\text{HHI}} \text{Deposit-HHI}_{b,t-1} + \gamma^{\text{HHI}} \Delta \text{FF}_t \times \text{Deposit-HHI}_{b,t-1} + \eta_{b,c} + \eta_{c,t} + \epsilon_{b,c,t}, \quad (\text{A2})$$

where $Y_{b,c,t}$ includes the change in deposit spread, growth rate of deposits, and mortgage and small business loan origination for bank b at quarter t . $\text{AdSpend}_{b,c,t-1}$ is lagged log values of advertising spending of bank b in county c in year $t-1$. ΔFF_t is the contemporaneous change in the Fed funds rate at t . $\text{Deposit-HHI}_{b,t-1}$ is the lagged bank-level deposit concentration for bank b in year $t-1$. $\eta_{b,c}$ and $\eta_{c,t}$ are bank-county and county-year fixed effects, respectively. We cluster standard errors at the county level.

Table A.7 presents the results on advertising spending, for bank liabilities (including the growth rate of deposits and change in deposit spread) and for bank assets (including mortgage and small business loan origination). On the liability side, column (1) shows that banks with higher advertising spending are able to charge higher deposit spreads when the Fed funds rate rises. At the same time, as column (2) indicates, banks that advertise more intensively experience larger deposit outflows. This is consistent with the prediction by the deposits channel of monetary policy (Drechsler et al., 2017), that banks with high market power experience larger increases in deposit spreads and greater deposit outflows during a Fed funds rate hike. On the asset side, columns (3) and (4) show that both mortgage and small business loan lending decline as the Fed funds rate rises. Here, as deposits are a special source of funding for banks, banks face a trade-off between maximizing profits from deposits and financing a large balance sheet. The combination of the findings from both the liability side and the asset side further supports the deposits channel.

Consider a bank (b)-county (c)-border ($d(c)$)-year (t) panel. Let $d(c)$ stand for the border group to which county c belongs. We estimate the causal effect of advertising on the demand for financial products using the following specifications:

$$Y_{b,c,t} = \beta \cdot \text{AdSpend or AdContent}_{b,c,t} + \eta_{b,c} + \eta_{b,d(c),t} + \epsilon_{b,c,d(c),t}, \quad (\text{A3})$$

where $Y_{b,c,t}$ is a set of outcome variables for various financial products, such as the growth rate of deposits, change in deposit spread, and mortgage and small business loan origination, of bank b in county c at t . Independent variables of interest can either be the advertising expenses or the bank images. $\ln(\text{Ad})_{b,c,t}$ is the log amount of product-specific advertising spending by b in county c at t , which is calculated according to which DMA county c belongs to. When examining differential effects of different bank images on demand for financial products, we replace the advertising variable with two measures of bank advertising content, including pricing information $1(\text{Pricing})_{b,c,t}$ and service quality $1(\text{Service})_{b,c,t}$, effectively omitting emotional appeal and trust building $1(\text{Emotion/Trust})_{b,c,t}$ which is used as the benchmark category.

Table A.8 presents the results of the border discontinuity design. Column (1) reveals a strong effect of deposit advertising on deposit growth rates. Specifically, a 1% increase in deposit advertising spending is associated with an additional 0.09 basis points increase in the deposit growth rate, relative to a sample mean of 3.75 percentage points. Column (2) indicates that banks with higher advertising expenditures tend to charge higher deposit spreads on average, aligning with prior evidence that advertising-intensive banks are able to maintain higher deposit spreads and market power. Columns (3) and (4) show that mortgage advertising has a significant positive effect on the dollar amount of mortgage origination and small business loan origination. Quantitatively, a one percent rise in mortgage advertising spending corresponds to a 0.01% increase in mortgage origination. As for small business loans, a one percent rise in advertising spending corresponds to a 0.007% increase in origination.

A.1.4. Financial Crisis

We study how macro shocks such as the Global Financial Crisis (GFC) alter bank advertising content. Using GFC exposure measure from Chodorow-Reich (2014), we define the treatment group of banks as those cosyndicated with Lehman Brothers more prior to the crisis. We estimate the specification below and compare advertising content from these “treated” banks relative to other banks operating in the same county.

$$\text{AdContent}_{b,c,t} = \beta \cdot \text{Post(GFC)}_t \times 1(\text{Lehman Cosyndication})_b + \eta_b + \eta_{c,t} + \epsilon_{b,c,t}. \quad (\text{A4})$$

Post(GFC)_t is a dummy variable that takes the value of 1 if the year t falls within the five-year window from 2007 to 2011, and the value of 0 otherwise. $1(\text{Lehman Cosyndication})_b$ is a dummy variable that takes the value of 1 if bank b has a high (i.e., above median) fraction of cosyndication loans where Lehman Brothers had a lead role, following Ivashina and Scharfstein (2010) and Chodorow-Reich (2014), and the value of 0 otherwise. As in Chodorow-Reich (2014), we focus on the 43 most active banks in the syndicated market for this analysis, resulting in a sample size of 31 banks for the final merged dataset.

Table A.9 shows the results. Banks more exposed to Lehman Brothers lean into providing pricing information and building trust. Intuitively, banks highlight pricing advantages as the shock likely weakens their market power. Banks also utilize the trust and emotion theme more in their advertising to counteract the rising distrust in the banking system, both during and after the financial crisis. This may help these banks mitigate the possibility of bank runs.

A.2. Appendix on Method

This appendix provides additional details of our method and presents additional analysis to validate the method and show its robustness.

A.2.1. Technical Details on the Unsupervised Video Embedding Method

1. Representation of Video Using High-Dimensional Multimodal Embedding.

The foundation of our method is the construction of video embeddings that capture the rich multimodal content of advertisements. For this purpose, we employ **OmniEmbed**,²⁰ a state-of-the-art open-source model built on Alibaba Cloud’s Qwen2.5-Omni-7B. **OmniEmbed** generates unified embeddings by jointly processing text, audio, images, and video inputs. This allows us to capture not only the rich content within the visual, verbal, and vocal dimensions of each advertisement, but also the interdependencies across these channels.

In practice, we first separate the original audio track and extract representative visual frames from the video using **moviepy**.²¹ We then apply the open-sourced speech-to-text model **Whisper**²² provided by OpenAI to convert the audio track into text. Since some advertisements are in non-English languages, we translate the transcripts into English using the Google Translate API through **deep-translator**.²³ The transcript, audio features, and visual frames are then provided simultaneously to **OmniEmbed**, which outputs a single 3,584-dimensional embedding for each advertisement.

2. Dimension Reduction Using PaCMAP. The raw video embeddings generated by **OmniEmbed** are 3,584-dimensional vectors, which are too high-dimensional for direct visualization or intuitive interpretation. To address this, we apply Pairwise Controlled Manifold Approximation Projection (**PaCMAP**),²⁴ a state-of-the-art nonlinear dimension reduction algorithm, to project these embeddings into a two-dimensional space. **PaCMAP** is designed to preserve both global and local structures of the data, meaning that points that are similar in the original high-

²⁰<https://huggingface.co/Tevatron/OmniEmbed-v0.1>

²¹<https://github.com/Zulko/moviepy>

²²<https://github.com/openai/whisper>

²³<https://github.com/nidhaloff/deep-translator>

²⁴<https://github.com/YingfanWang/PaCMAP>

dimensional space remain close in the reduced space, while broader clusters of dissimilar points are also kept well separated.

There are three key parameters in PaCMAP. The number of nearest neighbors, `n_neighbors`, controls the extent to which local structure is preserved by defining how many close neighbors are considered when constructing the low-dimensional representation. The ratio of the number of mid-near pairs to the number of neighbors, `MN_ratio`, adjusts the weight given to moderately close points, thereby helping to balance local and global geometry. The ratio of the number of further pairs to the number of neighbors, `FP_ratio`, determines the weight given to distant points, ensuring that broad cluster structures are maintained. In identifying the product categories, we set `n_neighbors` to 10, `MN_ratio` to 5, and `FP_ratio` to 20. In identifying the advertising content, we set `n_neighbors` to 10, `MN_ratio` to 10, and `FP_ratio` to 10. These parameters were chosen to optimize visualization clarity, and our framework and results are robust to alternative parameter choices.

By default, the PaCMAP algorithm applies a Principal Component Analysis (PCA) pre-processing step, reducing the original embeddings to 100 dimensions before applying the manifold learning procedure. This step substantially accelerates computation while retaining nearly all relevant information from the high-dimensional data.

3. Unsupervised Clustering Using Hierarchical Agglomerative Clustering. We next apply hierarchical agglomerative clustering (HAC) to the two-dimensional embeddings produced by PaCMAP. HAC is a bottom-up clustering method that recursively merges pairs of clusters based on a chosen linkage criterion. In our case, we use Euclidean distance together with Ward’s linkage method, which minimizes the increase in within-cluster variance at each step. This combination is widely used in the clustering literature and is particularly effective for producing compact, interpretable clusters.

A key advantage of HAC is that it naturally generates a hierarchy of clusters. This structure enables us to “cut” the tree at different levels, yielding clusterings of varying granularity. In our analysis, we cut the tree at the level of 30 clusters to obtain the most detailed product categories and advertising content. We then move to higher levels of the hierarchy to recover more aggregated clusters, which correspond to broader product categories and advertising

content.

While HAC provides only hard cluster assignments, advertisements may naturally overlap in advertising content (e.g., an advertisement can simultaneously emphasize pricing information and service quality). To account for this, we compute a soft assignment between each embedding and the cluster centroids. Following the soft clustering literature, we use the Student’s t-distribution as a kernel to measure the probability of assigning embedding $\tilde{d}_v$ to cluster $\tilde{\mu}_j$:

$$\Pr(v \in j) = \frac{\left(1 + \|\tilde{d}_v - \tilde{\mu}_j\|^2/\alpha\right)^{-\frac{\alpha+1}{2}}}{\sum_{j'} \left(1 + \|\tilde{d}_v - \tilde{\mu}_{j'}\|^2/\alpha\right)^{-\frac{\alpha+1}{2}}},$$

where α is the degrees of freedom of the Student’s t-distribution. We use $\alpha = 0.5$ to compute the soft assignment, and our results are robust to alternative choices of α . The resulting soft assignment can be interpreted as the probability of each video belonging to each cluster.

4. Interpretation of Clusters Using Generative Description and TF-IDF. To systematically interpret the clusters, we generate natural-language descriptions of each advertisement using Qwen2.5-Omni-7B, a state-of-the-art multimodal generative model. Each full video is provided to the model, including its transcript, audio track, and visual frames. We then prompt the model with the following instruction:

“Combining all the materials (including text, audio, and video) from the advertisement of financial products from a financial firm, describe this advertisement.”

The model outputs a concise description that summarizes the advertisement’s content in a standardized format. These generated descriptions provide consistency across ads, reduce noise from idiosyncratic transcripts or incidental details, and create a comparable text corpus for interpretation.

We next apply the term frequency-inverse document frequency (TF-IDF, Loughran and McDonald, 2011) method to the set of descriptions. Each video is represented as a TF-IDF vector, and the average of the TF-IDF vectors for all videos within it represents each cluster. The words with the highest average TF-IDF scores within each cluster serve as interpretable labels, highlighting the distinguishing features of each cluster.

A.2.2. Product Categorizations

As a preparation step, we identify the product categories using an unsupervised clustering approach. This is not directly related to quantifying advertising content delivered to customers, but this categorization is helpful in our analysis and within-product comparisons. The underlying intuition behind our product categorization is that the most significant variance in the embedding space likely reflects the core product features—that is, a mortgage advertisement is likely very different from a general banking service advertisement.

Our clustering method identifies five primary product categories: personal deposits, business banking, personal credit (including credit card, auto loan, and veteran finance), mortgages, investment and retirement services. These categories are well separated in the embedding space, as shown in Panel (a) of Figure A.5. A more granular level of clusters, shown in Panel (b) of Figure A.5, reveals thirty subcategories. For example, the personal credit category contains three sub-products, including credit card, auto loan, and veteran finance. We rely on the most important keywords to interpret the clusters. For example, the cluster labeled “personal deposit” is characterized by keywords such as “bank”, “checking account”, and “money”, while the “mortgage” cluster includes top keywords like “mortgage”, “home”, and “loan”.

A.2.3. Cross-Validation Using LLM

Cross-Validating Product Categories Using LLM We use a large language model (LLM) to identify the product categories by submitting the extracted transcript text to ChatGPT. Specifically, we ask ChatGPT to classify the text into one of the following categories: banking services, business loans, personal credit, mortgage and real estate, investment and brokerage, retirement financial services, and insurance. To ensure the reliability of GPT’s response, we fixed the random seed and set the temperature to 0, making GPT’s outputs more deterministic. We also ask GPT to provide the confidence level of its answer on a scale between 0 and 1 and an explanation for each answer, which allows for manual verification of the reasoning behind the scores.

In Panel (a) of Table A.3, we show that product categories extracted independently from the clustering method and LLM align very well. For example, among the videos categorized

as personal deposit by our clustering method, 70.8% are also categorized as personal deposit by LLM. Intuitively, the two methods should be very similar, as both are based on the exact text that has been extracted. This further validates the robustness of our results in extracting advertising categories.

Cross-Validating Advertising Content Quantification Using LLM Similarly, we also use the ChatGPT model to classify the text into the categories we obtained from the clustering method. In our prompt submitted to the model, we query the GPT with the name of the financial institution and the transcript of the advertisement, and in response, GPT generates a string of text containing the probability distribution of the list of categories in Table 2. In Panel (b) of Table A.3, we show that categorization extracted from the LLM is highly consistent with the clustering method.

A.2.4. More Examples of Advertising Content

In this subsection, we provide additional examples of how financial institutions shape their public image.

Example of Financial Information The financial information category intends to capture the financial benefits of the product, including interest rates, payment plans, and cashback rewards.

“While others just talk about their lending, we’re taking action. Right now, our fixed-rate commercial loans have rates as low as 3.5%. It’s time to get back to business.”

“Open a premium checking account and get \$300 when you set up direct deposit.”

Example of Service Quality The service quality category includes information on local and community service, online and mobile applications, and daily convenience.

“We’ll make your construction loan process quick and easy. And when you’re ready to move in, your final home loan can be ready, too.”

“A local bank that’s committed to our community and invested in its future. You’ll get access to banking experts, a partner in the mortgage process, convenient locations, an advisor for your business, and so much more. It all starts when you open your account.”

Example of Emotional Appeal and Trust Building Our last broad advertising content is the emotional appeal and trust building. For emotional appeal, we focus on the small business support, home and future, college and education, and sport sponsorship.

“At Edmonton Bank, we know the key role that small and medium-sized businesses play in the community. After all, we’ve been helping them grow for over a century. Our lenders have the expertise to help you make the right decision for you and your business. From financing a single store to multiple locations, you’ll always get competitive rates and friendly professional service that sets us apart from the competition.”

“Most ideas don’t begin in a bank. But we’ll join you wherever you break ground on your dreams. We exist where ideas exist. You have the dream. We have the team.”

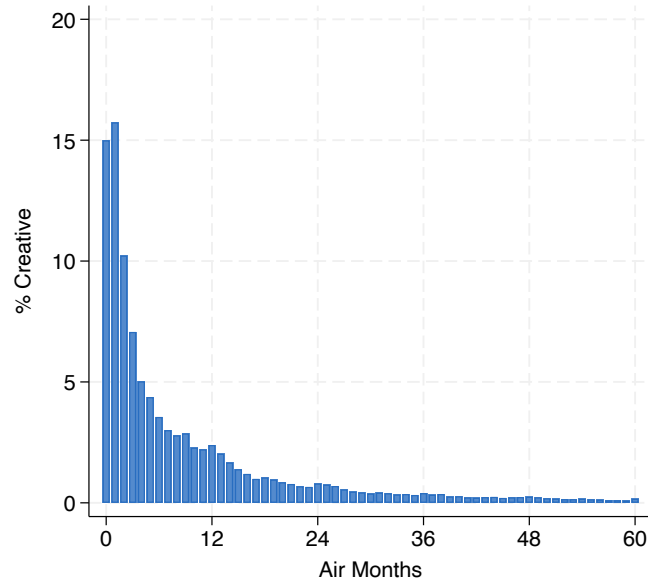
For trust building, we focus on the relationship building, such as brand promotion and home and community trust, and professional imagery, such as professional expertise and wealth management.

“We share our world with billions of people, each going through unique experiences, hopefully accomplishing just what they set out to do. But we can’t do it alone. We have a better chance of achieving our dreams with a great partner. For those times, trust us. Now in your corner and proud to be part of this community.”

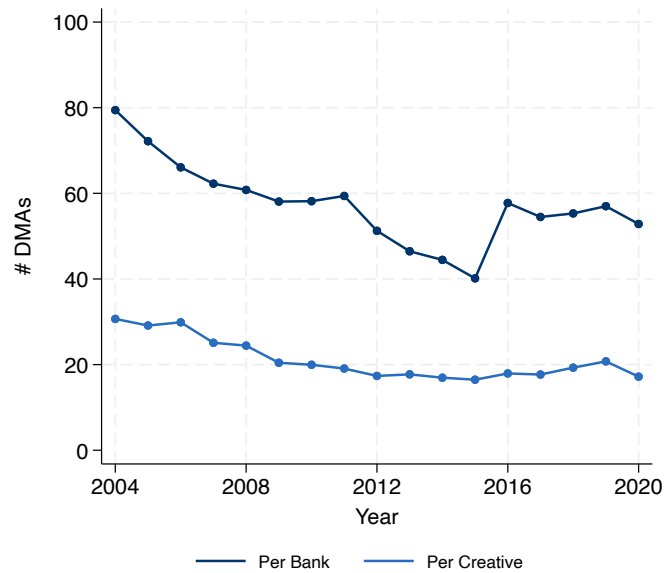
“Are you in good hands? At any minute, you could be a victim of fraud. Fraud could mean lower credit scores and higher interest rates when you apply for a credit card. It’s a problem waiting to happen. Check your credit score. ”

Figure A.1. Distribution of Air Months and Geographic Coverage of Financial Advertising

Panel (a): Air Months

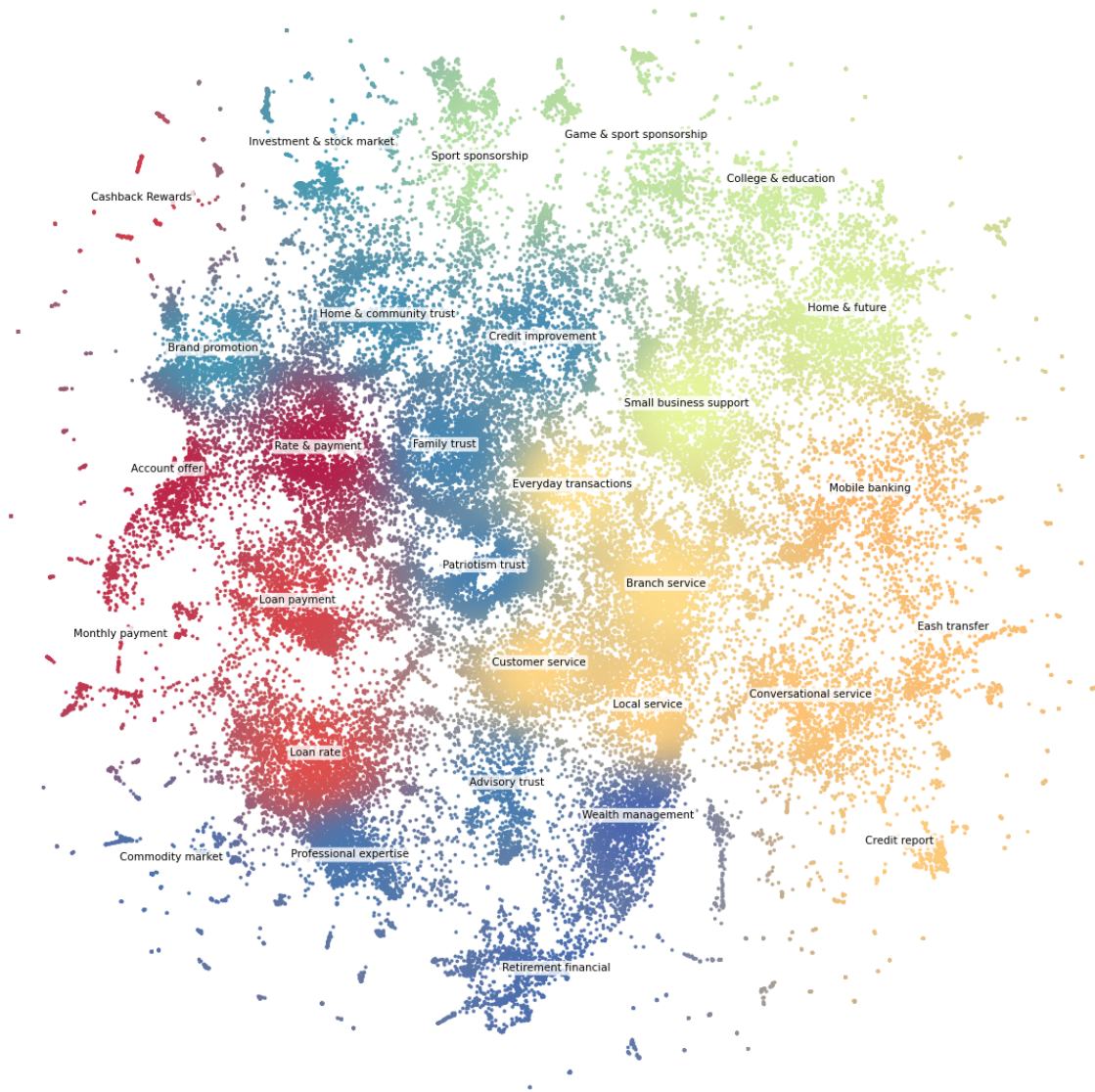


Panel (b): Geographic Coverage



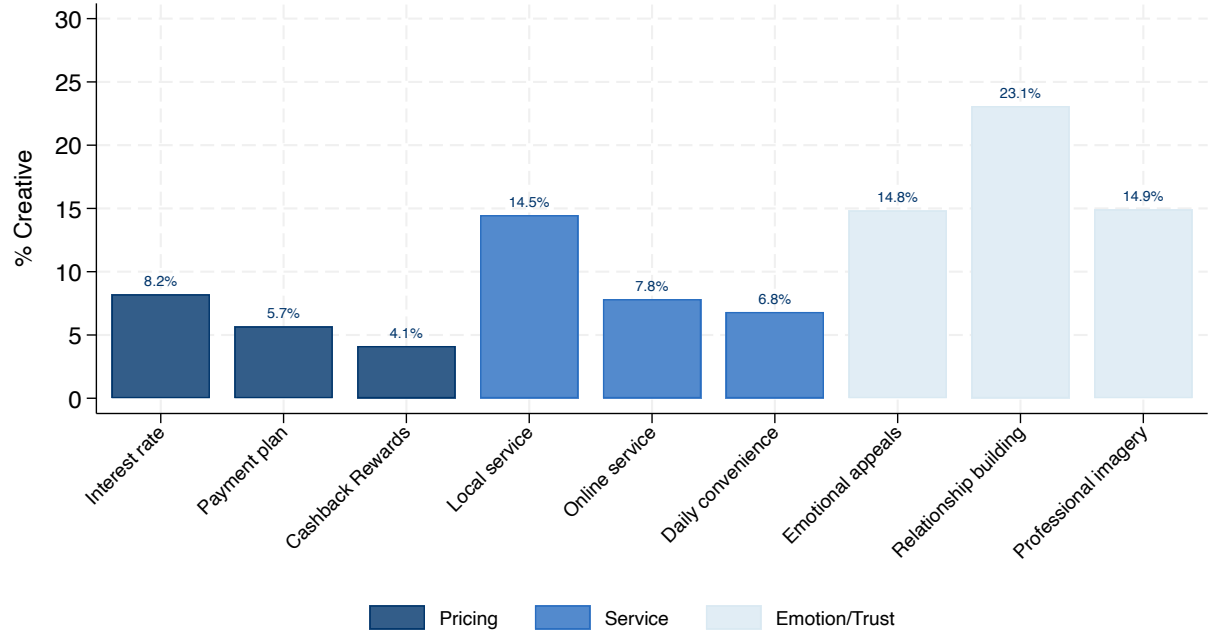
Notes. This figure shows the distribution of air months of financial advertising in Panel (a) and geographic coverage of financial advertising in Panel (b). In Panel (a), the tenure is measured in terms of the total number of months the advertising aired on television. In Panel (b), the dark blue line is the average number of DMAs where a bank has any financial advertising in a given year, and the light blue line is the average number of DMAs for a specific financial advertising.

Figure A.2. Visualization of the Video Embedding for Detailed Advertising Content



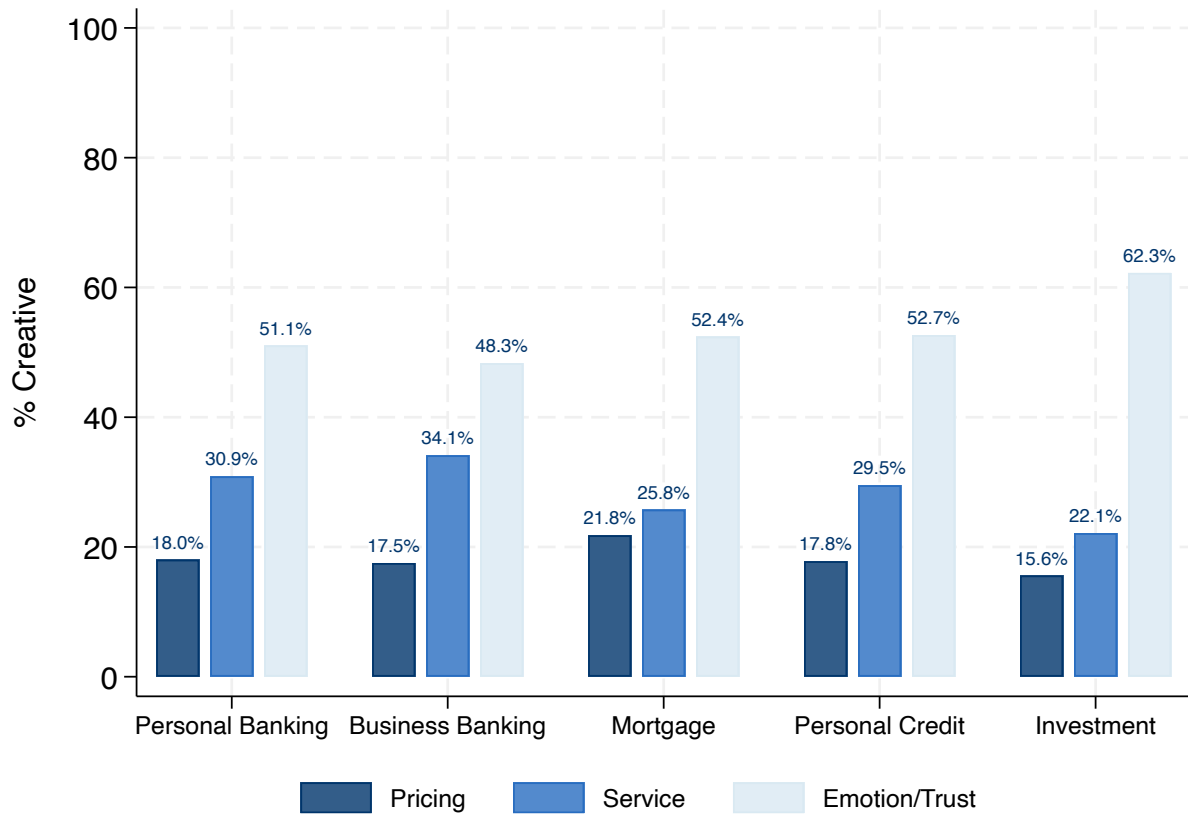
Notes. This figure shows the visualization of the 2-dimensional video embedding for detailed advertising content.

Figure A.3. Distribution of the Detailed Advertising Content



Notes. This figure shows the distribution of detailed advertising content for financial advertising. The list of financial advertising persuasive information can be found in Table 2. The financial information category contains detailed content including the interest rates, payment plans, and cashback rewards. The service quality category includes the local and community service, online and mobile applications, and daily convenience. The emotional appeal and trust building contains the emotional appeal, relationship building, and professional imagery.

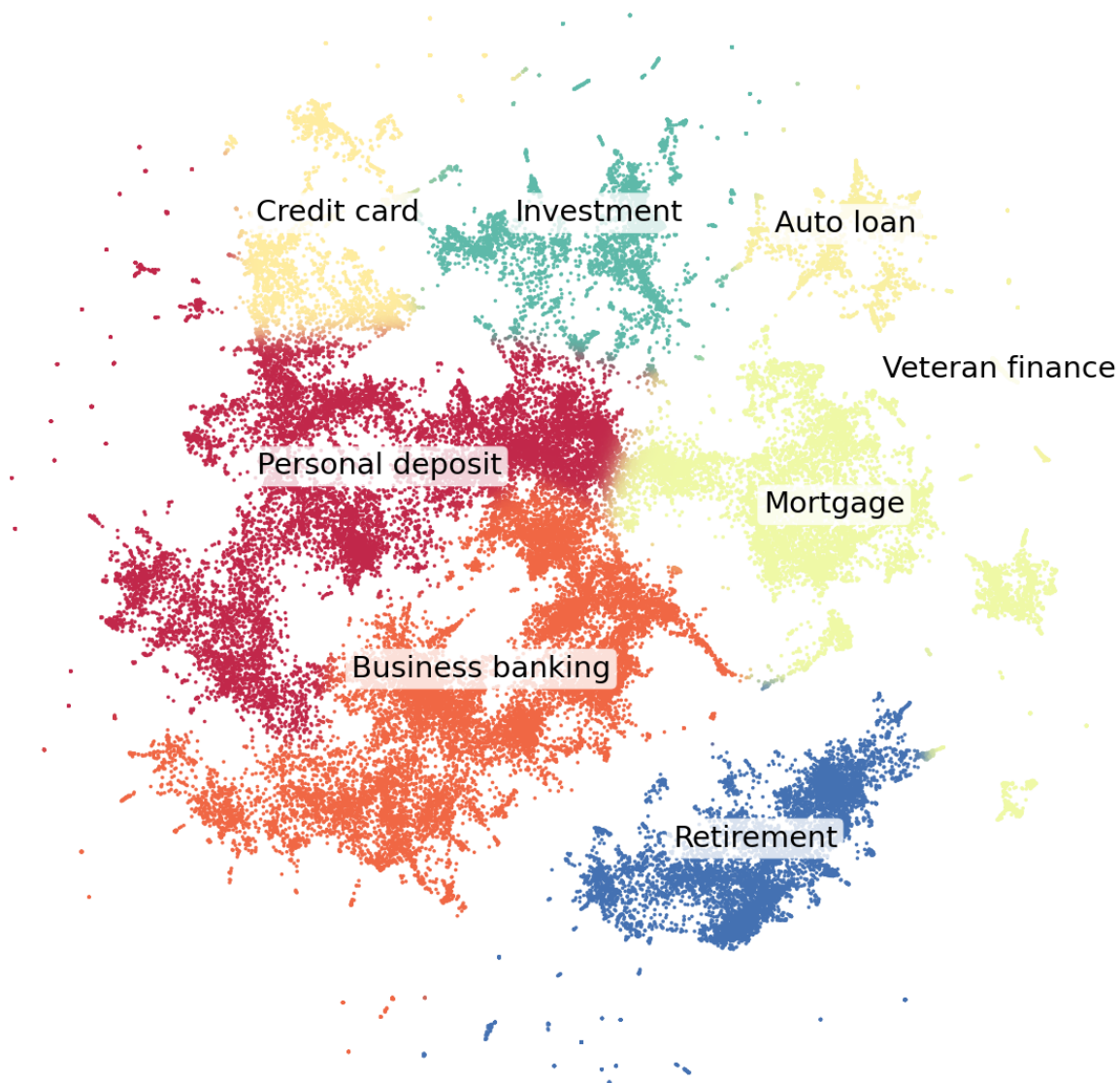
Figure A.4. Distribution of the Advertising Content by Product Categories



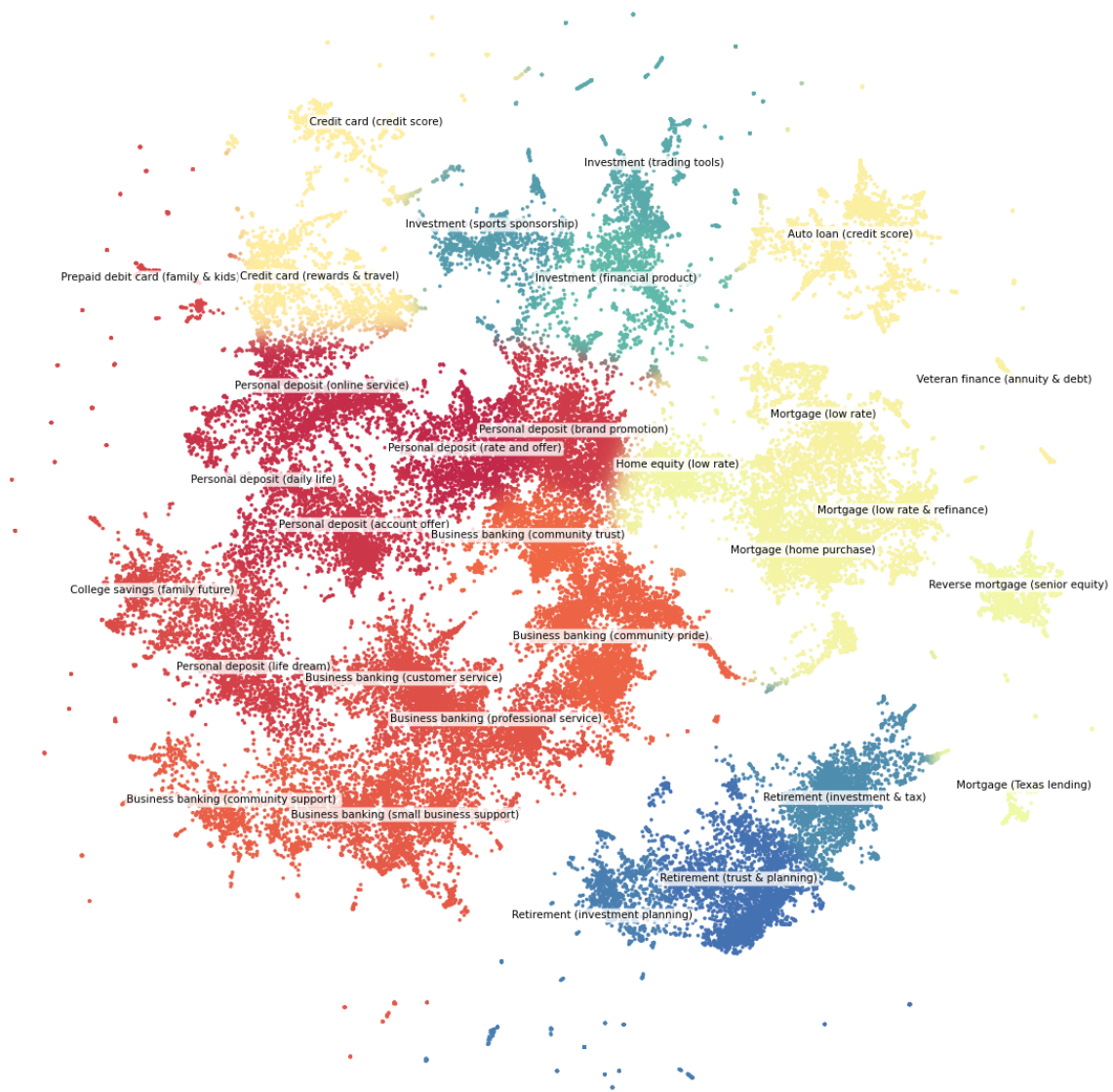
Notes. This figure shows the distribution of advertising content for different specific products. The list of financial advertising broad persuasive information can be found in Table 2.

Figure A.5. Visualization of the Video Embeddings for Product Category

Panel (a): Broader Product Category



Panel (b): Detailed Product Category



Notes. This figure shows the visualization of the two-dimensional video embeddings for broader product categories in Panel (a) and detailed product categories in Panel (b).

Table A.1. Bank Selection Into Advertising

	(1)	(2)	(3)	(4)
	$1(\text{Advertising})_{b,t}$			
$\ln(\text{Asset})_{b,t-1}$	0.028*** (0.010)			0.021 (0.037)
$\text{Z-Score}_{b,t-1}$	-0.002 (0.002)			-0.002 (0.002)
$\text{ROA}_{b,t-1}$	-0.059* (0.032)			-0.020 (0.024)
Tobin's $q_{b,t-1}$		-0.130* (0.073)		-0.125* (0.074)
Deposit Spread $_{b,t-1}$		-0.039** (0.016)		-0.026* (0.016)
Deposit Productivity $_{b,t-1}$		0.040** (0.019)		0.007 (0.068)
Asset Productivity $_{b,t-1}$		0.006 (0.012)		0.012 (0.012)
Deposits HHI $_{b,t-1}$			-0.905*** (0.313)	-0.913*** (0.309)
Mortgage HHI $_{b,t-1}$			-1.179 (0.734)	-1.078 (0.720)
Observations	12,125	12,125	12,125	12,125
R-squared	0.132	0.136	0.156	0.176
No. of Institutions	520	520	520	520
Mean of Y	0.729	0.729	0.729	0.729
Quarter FE	Y	Y	Y	Y

Notes. This table examines which banks select into advertising. The sample contains quarterly information of all the publicly listed U.S. bank holding companies. The outcome variable $1(\text{Advertising})_{b,t}$ is a dummy variable for whether bank b has non-missing and positive Compustat item Advertising and Marketing Expenses (`xadq`) in quarter t . The explanatory variables are lagged one period and include log total assets, Z-Score (the bank's ROA plus its equity ratio divided by its standard deviation of ROA), ROA, Tobin's q , deposit spread, deposit and asset productivity from Egan et al. (2022), and the bank's deposit and mortgage market concentration HHI, calculated as the volume-weighted average of the county-level HHI across all the counties where the bank operates. Standard errors are double-clustered at the levels of bank and quarter. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table A.2. Markov Transition Matrix of Advertising Content

Panel (a): For All Financial Advertising

AdContent _{<i>b,t</i>}	AdContent _{<i>b,t+1</i>}		
	Pricing	Service	Emotion/Trust
Pricing	22.08%	31.63%	46.28%
Service	4.16%	64.37%	31.47%
Emotion/Trust	6.40%	34.97%	58.63%

Panel (b): For Financial Advertising Related to Deposits

AdContent _{<i>b,t</i>}	AdContent _{<i>b,t+1</i>}		
	Pricing	Service	Emotion/Trust
Pricing	22.91%	30.45%	46.65%
Service	3.70%	68.33%	27.97%
Emotion/Trust	6.61%	34.64%	58.75%

Panel (c): For Financial Advertising Related to Loans

AdContent _{<i>b,t</i>}	AdContent _{<i>b,t+1</i>}		
	Pricing	Service	Emotion/Trust
Pricing	31.47%	25.10%	43.43%
Service	3.04%	73.60%	23.36%
Emotion/Trust	6.33%	31.78%	61.89%

Notes. This table presents the Markov transition matrix of advertising content over year. Panel (a) contains advertising content at the bank-year level using all the financial advertising. Panel (b) and Panel (c) contains advertising content at the bank-year level using financial advertising related to deposits and loans, respectively. To estimate the transition matrix, we only consider banks with at least 5 year sample of financial advertising data.

Table A.3. Alignment of Product Categories and Advertising Content Between Clustering Method and LLM

Panel (a) Product Category

	Product Category (Clustering)				
	Personal deposit	Business banking	Personal credit	Mortgage	Investment/retirement
# Videos	13,746	16,410	5,466	10,093	11,866
<i>Distribution of Product Categories Through LLM</i>					
% Banking	0.708	0.737	0.227	0.185	0.126
% Business loan	0.017	0.066	0.009	0.006	0.006
% Personal credit	0.101	0.042	0.568	0.049	0.035
% Mortgage	0.044	0.048	0.050	0.692	0.027
% Investment and brokerage	0.096	0.084	0.107	0.044	0.517
% Retirement financial services	0.026	0.014	0.028	0.019	0.265
% Insurance	0.008	0.009	0.009	0.004	0.024
<i>Distribution of Product Categories Through LLM (Standardized)</i>					
% Banking	0.625	0.698	-0.582	-0.688	-0.836
% Business loan	-0.076	0.340	-0.141	-0.168	-0.174
% Personal credit	-0.019	-0.261	1.887	-0.231	-0.289
% Mortgage	-0.338	-0.324	-0.318	1.613	-0.387
% Investment and brokerage	-0.253	-0.294	-0.216	-0.428	1.164
% Retirement financial services	-0.229	-0.287	-0.215	-0.260	0.982
% Insurance	-0.038	-0.031	-0.018	-0.095	0.176

Panel (b) Product Category

	Advertising Content (Clustering)			
	Pricing	Service	Emotion	Trust
# Videos	10,535	16,585	8,512	21,948
<i>Distribution of Advertising Content Through LLM</i>				
% Pricing	0.271	0.211	0.190	0.220
% Service	0.378	0.420	0.354	0.350
% Emotion	0.077	0.086	0.139	0.086
% Trust	0.273	0.282	0.317	0.344
<i>Distribution of Advertising Content Through LLM (Standardized)</i>				
% Pricing	0.166	-0.039	-0.110	-0.008
% Service	0.007	0.140	-0.069	-0.083
% Emotion	-0.075	-0.030	0.228	-0.030
% Trust	-0.098	-0.073	0.021	0.094

Notes. This table compares the alignment of product categories and advertising content obtained specifically through the clustering method and LLM. Panel (a) and Panel (b) shows the distribution of product categories and advertising content, respectively. In each panel, the standardized distribution is computed by subtracting the mean and dividing by the standard deviation of the distribution.

Table A.4. Advertising Content by Service Quality Banks: Alternative Measure

	(1)	(2)	(3)
Measure of Service Quality	Branch Network		
Ad Category	Deposit		
% AdContent _{<i>b,c,t</i>}	Pricing	Service	Emotion/Trust
High Service Quality _{<i>b,c,t-1</i>}	-0.748*** (0.042)	0.523*** (0.091)	0.226*** (0.069)
Observations	987,169	987,169	987,169
R-squared	0.053	0.059	0.060
No. of Institutions	1,120	1,120	1,120
Mean of Y	13.30	36.41	50.28
County-Year FE	Y	Y	Y

Notes. This table examines the advertising content across banks with different levels of service quality in a bank(*b*)-county(*c*)-year(*t*) panel. The analysis considers the deposit business. % AdContent_{*b,c,t*} is the share of advertising content used by bank *b* in county *c* in year *t*. High Service Quality_{*b,c,t-1*} is a dummy variable for whether bank *b* is above the 80 percent quantiles of service quality among all the banks in the county *c* in year *t* - 1, using number of branches scaled by deposits as the measure of service quality. The model includes county-year fixed effects. Standard errors are clustered at the county level. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table A.5. Advertising Content by High Versus Low Local Market Shares

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Product Market	SOD Deposit			HMDA Mortgage			CRA Loan		
Ad Category	Deposit			Mortgage			Loan		
% AdContent _{b,c,t}	Pricing	Service	Emotion/Trust	Pricing	Service	Emotion/Trust	Pricing	Service	Emotion/Trust
Large (Market Share) _{b,c,t-1}	-0.412*** (0.072)	2.506*** (0.117)	-2.095*** (0.097)	-1.257*** (0.037)	-0.286*** (0.062)	1.542*** (0.059)	-0.293*** (0.052)	-0.284*** (0.095)	0.577*** (0.079)
Observations	922,811	922,811	922,811	706,796	706,796	706,796	520,831	520,831	520,831
R-squared	0.054	0.068	0.068	0.064	0.067	0.067	0.075	0.087	0.096
No. of Institutions	937	937	937	721	721	721	425	425	425
Mean of Y	13.44	36.16	50.40	14.51	34.60	50.89	13.72	36.72	49.56
County-Year FE	Y	Y	Y	Y	Y	Y	Y	Y	Y

Notes. This table examines the advertising content across banks with different levels of local market power in a bank(b)-county(c)-year(t) panel. The analysis considers three different products, including deposits in Columns (1)-(3), mortgages in Columns (4)-(6), and business loans in Columns (7)-(9). % AdContent_{b,c,t} is the share of advertising content for that specific product used by bank b in county c in year t . Large (Market Share)_{b,c,t-1} is a dummy variable for whether bank b is above the 80 percent quantiles of market shares in the county c in the respective product market in year $t-1$. The model includes county-year fixed effects. Standard errors are clustered at the county level. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table A.6. Advertising Spending and Bank Value

	(1)	(2)	(3)	(4)	(5)
$Y_{b,t}$	Deposit TFP	Asset TFP	Deposit Spread	Deposit Beta	Tobin's q
$Ad_{b,t-1}$	0.012* (0.006)	0.063*** (0.013)	0.073*** (0.015)	0.001 (0.002)	0.022*** (0.004)
Observations	11,930	11,930	19,800	16,145	19,410
R-squared	0.955	0.207	0.493	0.102	0.474
No. of Institutions	481	481	700	472	692
Mean of Y	0.0323	0.0413	0.452	0.420	1.203
Bank Controls	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y

Notes. This table examines the relationship between bank advertising spending and bank value in a bank(b)-quarter(t) panel. The sample contains all the publicly listed U.S. bank holding companies with non-missing and positive Advertising and Marketing Expenses (Compustat item `xadq`). The outcome variables are bank deposit productivity (Deposit TFP) and asset productivity (Asset TFP) in Columns (1)-(2) from [Egan et al. \(2022\)](#), deposit spread and deposit beta in Columns (3)-(4) from [Drechsler et al. \(2021\)](#), and Tobin's q . $Ad_{b,t-1}$ is Compustat advertising expense scaled by Call Reports total non-interest expense for bank b in quarter $t - 1$. The model includes quarter fixed effects and lagged bank controls. Lagged bank controls include the log of the bank's total assets, the bank's Z-Score (the bank's ROA plus its equity ratio divided by its standard deviation of ROA), the bank's ROA, the bank's equity to assets ratio, the bank's deposits to assets ratio, and the bank's deposit and mortgage market concentration HHI, following [Gelman et al. \(2023\)](#) and [Egan et al. \(2022\)](#). Standard errors are clustered at the bank level with a wild cluster bootstrap procedure (1000 times). *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table A.7. Advertising Spending and Monetary Policy Transmission

	(1)	(2)	(3)	(4)
$Y_{b,c,t}$	$\Delta(\text{Deposit Spread})$	$\Delta \ln(\text{Deposits})$	$\ln(\text{Mortgage})$	$\ln(\text{Small Business Loan})$
$\ln(Ad)_{b,c,t-1}$	0.0004 (0.0004)	0.0001 (0.0002)	0.0064*** (0.0018)	0.0009 (0.0008)
$\Delta FF_t \times \ln(Ad)_{b,c,t-1}$	0.0086*** (0.0007)	-0.0002* (0.0001)	-0.0077*** (0.0012)	-0.0042*** (0.0007)
Observations	51,516	86,829	684,850	485,658
R-squared	0.915	0.491	0.643	0.830
No. of Institutions	603	809	1,019	351
Mean of Y	0.140	0.0567	11.89	12.48
$HHI_{b,t-1}$ and $\Delta FF_t \times HHI_{b,t-1}$	Y	Y	Y	Y
County-Year FE	Y	Y	Y	Y
Bank-County FE	Y	Y	Y	Y

Notes. This table examines the relationship between advertising spending and the transmission of monetary policy in a bank(b)-county(c)-year(t) panel. $\ln(Ad)_{b,c,t-1}$ is the lagged log advertising spending for bank b in county c in year $t - 1$. ΔFF_t is the contemporaneous change in the Fed funds rate. Following [Drechsler et al. \(2017\)](#), we control for bank-level Deposit-HHI $_{b,t-1}$ and its interaction term $\Delta FF_t \times HHI_{b,t-1}$. Bank-level Deposit-HHI $_{b,t-1}$ is calculated as the deposit-weighted average of the county-level HHI across all the counties where the bank b operates in year $t - 1$. The model includes county-year fixed effects and bank-county fixed effects. Standard errors are clustered at the county level. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table A.8. Advertising Spending and Financial Product Demand:
Border Discontinuity Design

Ad Category $Y_{b,c,t}$	(1)	(2)	(3)	(4)
	Deposit		Mortgage	Loan
	$\Delta \ln(\text{Deposits})$	$\Delta(\text{Deposit Spread})$	$\ln(\text{Mortgage})$	$\ln(\text{Small Business Loan})$
$\ln(\text{Ad})_{b,c,t}$	0.0009** (0.0004)	0.0017*** (0.0006)	0.010** (0.004)	0.007*** (0.001)
Observations	48,709	48,709	335,171	401,296
R-squared	0.284	0.954	0.680	0.784
No. of Institutions	352	352	930	418
Mean of Y	0.0375	-0.0072	11.64	12.35
Bank-County FE	Y	Y	Y	Y
Bank-Border-Year FE	Y	Y	Y	Y

Notes. This table examines the relationship between advertising spending and demand for financial products using a border discontinuity design in a bank(b)-county(c)-border($d(c)$)-year(t) panel. The outcome variables, $Y_{b,c,t}$, include the changes in the log of deposits $\Delta \ln(\text{Deposits})$ in Column (1), changes in the deposit spread $\Delta(\text{Deposit Spread})$ in Column (2), the log amount of new mortgage origination $\ln(\text{Mortgage})$ in Column (3), and the log amount of new small business loan origination $\ln(\text{Small Business Loan})$ from CRA in Column (4). $\ln(\text{Ad})_{b,c,t}$ is the log amount of advertising spending for bank b in county c in year t . We use product-specific advertising measures, namely deposits in Column (1)-(2), mortgages in Column (3), and business loans in Column (4). The model includes bank-county fixed effects and bank-border-year fixed effects. Standard errors are clustered at the bank-county level and border-year level. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table A.9. Advertising Content After Economic Shocks: Financial Crisis

Ad Category % AdContent $_{b,c,t}$	(1)	(2)	(3)
	All		
	Pricing	Service	Emotion/Trust
$\text{Post}(\text{GFC})_t$	1.680***	-5.236***	3.556***
$\times 1(\text{Lehman Cosyndication})_b$	(0.069)	(0.136)	(0.117)
Observations	502,925	502,925	502,925
R-squared	0.194	0.163	0.150
No. of Institutions	31	31	31
Mean of Y	12.95	37.94	49.11
Bank FE	Y	Y	Y
County-Year FE	Y	Y	Y

Notes. This table examines how bank advertising content changes after financial crisis, using a bank(b)-county(c)-year(t) panel. $\text{Post}(\text{GFC})_t$ is a dummy variable for whether year t falls in the range of 2007 to 2011. $1(\text{Lehman Cosyndication})_b$ is a dummy variable for whether bank b had an above-median share of cosyndication loans where Lehman Brothers had a lead role, following Chodorow-Reich (2014). Fixed effects are denoted at the bottom. Standard errors are clustered at the county level. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table A.10. Balance Test: Border Discontinuity Design

	(1) ln(Ad)	(2) Pricing	(3) Service	(4) Emotion/Trust
ln(Income)	0.0116 (0.0585)	0.0009 (0.0021)	−0.0024 (0.0035)	0.0015 (0.0029)
ln(Population)	−0.0380 (0.1073)	−0.0059* (0.0034)	0.0031 (0.0059)	0.0027 (0.0046)
Female (%)	0.0024 (0.0073)	0.0001 (0.0003)	−0.0004 (0.0004)	0.0002 (0.0003)
Minority (%)	−0.0026 (0.0037)	0.0001 (0.0001)	−0.0003 (0.0002)	0.0001 (0.0002)
Age<20 (%)	−0.0043 (0.0053)	−0.0000 (0.0002)	0.0003 (0.0003)	−0.0003 (0.0002)
20<=Age<35 (%)	−0.0029 (0.0060)	−0.0001 (0.0002)	0.0002 (0.0004)	−0.0001 (0.0003)
35<=Age<60 (%)	−0.0036 (0.0051)	0.0000 (0.0002)	0.0001 (0.0003)	−0.0002 (0.0002)
Less than High School (%)	0.0403 (0.3222)	0.0086 (0.0143)	−0.0304 (0.0219)	0.0218 (0.0159)
High School (%)	0.1464 (0.2480)	0.0025 (0.0112)	−0.0134 (0.0172)	0.0110 (0.0126)
College (%)	−0.0298 (0.2613)	0.0015 (0.0117)	−0.0136 (0.0182)	0.0121 (0.0134)
Observations	2,814,698	2,814,698	2,814,698	2,814,698
R-squared	0.636	0.269	0.279	0.262
No. of Institutions	2,361	2,361	2,361	2,361
Mean of Y	4.642	0.140	0.366	0.494
Bank-Border-Year FE	Y	Y	Y	Y
Bank-County FE	Y	Y	Y	Y

Notes. This table examines the balance test across the borders using a border discontinuity design in a bank(*b*)-county(*c*)-border(*d*)-year(*t*) panel. The outcome variables include the log amount of advertising spending and the three categories of advertising content. The model includes bank-county fixed effects and bank-border-year fixed effects. Standard errors are clustered at the bank-county level and border-year level. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

A.3. Technical Details for the Conceptual Framework

A.3.1. More Details on Rate Awareness

Bank-specific rate awareness is a general function of pricing content:

$$\bar{\alpha}_j = \alpha(a_j^P, a_{-j}^P), \quad (\text{A5})$$

with partial derivatives $\alpha_1 \equiv \partial\alpha/\partial a_j^P > 0$, $\alpha_2 \equiv \partial\alpha/\partial a_{-j}^P > 0$, and $\alpha_1 > \alpha_2$. We also assume $\alpha(\cdot) > 0$ throughout (depositors always have some rate awareness).

Own pricing content raises $\bar{\alpha}_j$ by α_1 , which is larger than the cross-bank effect α_2 . The difference $\alpha_1 - \alpha_2 > 0$ captures the direct effect of own-bank pricing content on its depositors' rate focus, beyond the market-level broadcast channel on the rival. The broadcast channel (α_2) is symmetric: each bank's pricing content raises rate awareness for all depositors in the DMA.

In the between-bank comparison:

$$\bar{\alpha}_H - \bar{\alpha}_L = \alpha(a_H^{P*}, a_L^{P*}) - \alpha(a_L^{P*}, a_H^{P*}). \quad (\text{A6})$$

Since the function α is more responsive to its first argument than its second ($\alpha_1 > \alpha_2$), the bank with lower pricing content has lower rate awareness. Formally, by symmetry of the evaluation points (a_H^P, a_L^P) vs. (a_L^P, a_H^P) , the difference can be written as a line integral:

$$\bar{\alpha}_H - \bar{\alpha}_L = (a_H^{P*} - a_L^{P*}) \int_0^1 [\alpha_1(\mathbf{a}(t)) - \alpha_2(\mathbf{a}(t))] dt, \quad (\text{A7})$$

where $\mathbf{a}(t) = (1-t)(a_H^{P*}, a_L^{P*}) + t(a_L^{P*}, a_H^{P*})$ is a path between the two evaluation points.²⁵ Since $a_H^{P*} < a_L^{P*}$ (Proposition 1 (i)) and $\alpha_1 > \alpha_2$ everywhere, we have $\bar{\alpha}_H < \bar{\alpha}_L$: Bank H 's depositors are sleepier.

The specification is motivated by broadcast television advertising: pricing content from any bank raises rate awareness for all depositors in the DMA, but a bank's own pricing content has a larger effect on its own depositors because it directly orients them toward rate

²⁵To see this, define $g(t) = \alpha((1-t)a_H^{P*} + ta_L^{P*}, (1-t)a_L^{P*} + ta_H^{P*})$. Then $g(0) = \bar{\alpha}_H$, $g(1) = \bar{\alpha}_L$, and $g'(t) = (a_L^P - a_H^P)\alpha_1(\mathbf{a}(t)) + (a_H^P - a_L^P)\alpha_2(\mathbf{a}(t)) = (a_L^P - a_H^P)[\alpha_1(\mathbf{a}(t)) - \alpha_2(\mathbf{a}(t))]$. Integrating gives (A7).

comparisons. The own-vs.-cross asymmetry ($\alpha_1 > \alpha_2$) ensures that banks can differentially influence their own depositors' rate attention through content choice. Our results require only the three derivative conditions, not a specific functional form. A tractable parametrization that we use for illustrative purposes (e.g., the calibration in Appendix A.3.4) is the linear specification:

$$\bar{\alpha}_j = \alpha_0 + \alpha_1 a_j^P + \alpha_2 a_{-j}^P, \quad \alpha_0 > 0, \quad \alpha_1 > \alpha_2 > 0, \quad (\text{A8})$$

where α_0 is the baseline rate awareness when neither bank produces pricing content. All propositions hold for any smooth $\alpha(\cdot)$ satisfying the three derivative conditions; we invoke the linear form only when explicit closed-form expressions or numerical illustrations are needed.

This formulation is grounded in two recent empirical contributions. Egan et al. (2025) document that depositors infrequently shop for deposits, with 5–15% opening a new account each year, and model depositors as alternating between “awake” and “asleep” states following Beggs and Klemperer (1992) and Brown et al. (2023). They estimate that depositor sleepiness accounts for 58% of the average bank's deposit franchise value. Lu and Wu (2026) use transaction-level data on millions of U.S. depositors to show that inattention varies widely: unscheduled income remains in low-rate accounts much longer than scheduled income, and banks serving more inattentive depositors set lower rates and have weaker pass-through.

A.3.2. Proof of Proposition 1

Bank j maximizes its profit:

$$\max_{\rho_j, a_j^S, a_j^T} \Pi_j = \rho_j D_j, \quad \text{s.t. } a_j^P + a_j^S + a_j^T = 1. \quad (\text{A9})$$

where logit demand is $D_j = \exp(u_j)/[\exp(u_H) + \exp(u_L) + 1]$. The depositor's mean indirect utility is $u_j = -\bar{\alpha}_j \rho_j + A_j$, where $A_j = [(q_j a_j^S)^{1-1/\eta} + (\xi a_j^T)^{1-1/\eta}]^{\eta/(\eta-1)}$, and $\bar{\alpha}_j = \alpha(a_j^P, a_{-j}^P)$ with $\alpha_1 \equiv \partial\alpha/\partial a_j^P > \alpha_2 \equiv \partial\alpha/\partial a_{-j}^P > 0$.

First-order conditions for spread. The standard logit demand gives the rate FOC:

$$\rho_j^* = \frac{1}{\bar{\alpha}_j(1 - D_j)}. \quad (\text{A10})$$

First-order conditions for content. Using the envelope theorem at $\rho_j = \rho_j^*$, the content FOCs are as follows. At fixed a_j^P (so $da_j^T = -da_j^S$ and $\bar{\alpha}_j$ is unchanged), equating the marginal value of service and trust gives the composition FOC:

$$q_j^{1-1/\eta} (a_j^{S*})^{-1/\eta} = \xi^{1-1/\eta} (a_j^{T*})^{-1/\eta}, \quad (\text{A11})$$

which solves to:

$$\frac{a_j^{S*}}{a_j^{T*}} = \left(\frac{q_j}{\xi} \right)^{\eta-1}. \quad (\text{A12})$$

Note that since $\bar{\alpha}_j$ depends on a_j^P (not the composition of service and trust), the rate-sensitivity terms cancel in the service-vs.-trust allocation, and therefore the equilibrium composition is pinned down by the CES bundling.

The marginal demand from increasing a_j^S (reducing a_j^P , at fixed a_j^T) is:

$$\frac{\partial D_j}{\partial a_j^S} = D_j \left[\underbrace{\left(\alpha_1 \rho_j^* + \frac{\partial A_j}{\partial a_j^S} \right) (1 - D_j)}_{\text{own benefit}} - \underbrace{D_{-j} \alpha_2 \rho_{-j}^*}_{\text{cross-bank cost}} \right], \quad (\text{A13})$$

where $\partial A_j / \partial a_j^S = q_j (q_j a_j^S / A_j)^{-1/\eta}$. The own benefit has two components: the rate-sensitivity reduction ($\alpha_1 \rho_j^*$: reducing $\bar{\alpha}_j$ is valued at the current spread) and the non-price value gain from advertising consumption ($\partial A_j / \partial a_j^S$). The cross-bank cost: increasing a_j^S lowers a_j^P , which also reduces $\bar{\alpha}_{-j}$ by α_2 , benefiting the rival.

Setting the bracketed expression to zero gives the pricing FOC:

$$\left[\alpha_1 \rho_j^* + \frac{\partial A_j}{\partial a_j^S} \right] (1 - D_j) = D_{-j} \alpha_2 \rho_{-j}^*. \quad (\text{A14})$$

The system has four independent equations (two rate FOCs, plus for each bank one content condition) in four unknowns ($\rho_H, \rho_L, a_H^P, a_L^P$), with the a_j^S - a_j^T split determined by the

composition condition (A12) and the budget constraint.

Interior solutions. Here we show that interior solutions arise from two economic forces.

Interior service and trust (CES Inada condition). Fixing a_j^P constant, the marginal (net) benefit of shifting from trust to service is proportional to $\partial A_j / \partial a_j^S - \partial A_j / \partial a_j^T$ (the $\bar{\alpha}_j$ terms cancel since they depend only on a_j^P , not the service-trust split). As $a_j^S \rightarrow 0$: $(a_j^S)^{-1/\eta} \rightarrow +\infty$; the bank wants more service. As $a_j^T \rightarrow 0$: $(a_j^T)^{-1/\eta} \rightarrow +\infty$; the bank wants more trust. A unique interior $a_j^{S*} \in (0, 1 - a_j^P)$ exists by continuity.

Interior pricing (cross-bank externality). At $a_j^P = 1$ ($a_j^S, a_j^T \rightarrow 0$): the CES Inada terms dominate; the bank strictly gains from any additional non-price content. At $a_j^P = 0$: the CES terms are finite while the cross-bank cost $D_{-j}\alpha_2\rho_{-j}^*$ is strictly positive. Interior pricing obtains when the cross-bank externality is sufficiently strong: informally, α_2 must be large enough relative to the own-bank benefit (governed by α_1 and $A^{\max}$) that each bank finds it unprofitable to eliminate pricing content entirely.²⁶ This ensures the cross-bank cost dominates at $a_j^P = 0$, so $a_j^{P*} \in (0, 1)$.

Part (i): $a_H^{P} < a_L^{P*}$.* We establish that Bank H 's optimal pricing content is lower than Bank L 's by showing the marginal return to non-price content exhibits increasing differences in quality.

Step 1: The marginal return to non-price content is increasing in q_j . At fixed rival strategy, bank j 's marginal profit from increasing a_j^S (reducing a_j^P) is $\rho_j^* \partial D_j / \partial a_j^S$, which from (A13) equals:

$$\rho_j^* D_j \left[(\alpha_1 \rho_j^* + \partial A_j / \partial a_j^S)(1 - D_j) - D_{-j} \alpha_2 \rho_{-j}^* \right].$$

The only term that depends directly on q_j is $\partial A_j / \partial a_j^S = q_j (q_j a_j^S / A_j)^{-1/\eta}$, which is strictly increasing in q_j for any interior allocation (holding a_j^S fixed). The cross-bank cost $D_{-j} \alpha_2 \rho_{-j}^*$ depends on the rival's strategy, not on own quality. Through general equilibrium, higher q_j

²⁶For the linear specification $\bar{\alpha}_j = \alpha_0 + \alpha_1 a_j^P + \alpha_2 a_{-j}^P$, a sufficient condition is $\alpha_2 > 4(A^{\max} + \alpha_0)$ where $A^{\max} = \max_j [(q_j)^{1-1/\eta} + \xi^{1-1/\eta}]^{\eta/(\eta-1)}$. The bound follows from comparing the upper bound on the own benefit at $a_j^P = 0$ (at most $\alpha_1/\alpha_0 + A^{\max}$, using $\rho_j^*(1 - D_j) = 1/\bar{\alpha}_j \leq 1/\alpha_0$) with the lower bound on the cross-bank cost ($\geq \alpha_2/(4\bar{\alpha}_{-j})$, using $D_{-j}(1 - D_{-j}) \leq 1/4$).

also raises D_j (higher utility attracts more depositors), which raises $\rho_j^* = 1/[\bar{\alpha}_j(1 - D_j)]$ and lowers D_{-j} —both effects further increase the marginal return to non-price content.

Step 2: Bank H's best-response a_H^{P} is lower.* Consider both banks facing a common rival content level a_{-j}^P . Since the marginal return to reducing a_j^P is increasing in q_j , and $q_H > q_L$, bank H has a strictly stronger incentive to reduce pricing content at any common allocation. By monotone comparative statics, bank j 's best-response $a_j^{p, BR}(a_{-j}^P)$ is decreasing in q_j : $a_H^{p, BR}(a_L^P) < a_L^{p, BR}(a_H^P)$ for any rival strategy.

Step 3: Threshold existence. When q_H/q_L is close to 1, the quality advantage may be too small to overcome the equilibrium feedback. But as $q_H/q_L \rightarrow \infty$, bank H 's marginal return to service content grows without bound (since $\partial A_H / \partial a_H^S \propto q_H^{1-1/\eta} \rightarrow \infty$), while the cross-bank cost is bounded (since $D_{-j}\alpha_2\rho_{-j}^* \leq \alpha_2 / \inf \bar{\alpha}_{-j}$). Hence there exists a finite threshold $\theta^* > 1$ such that for $q_H/q_L > \theta^*$, bank H 's best response has strictly lower a_j^P than bank L 's for *all* possible rival strategies, ensuring $a_H^{P*} < a_L^{P*}$ in any equilibrium. $\square$

Part (ii): Service share increasing in quality. From (A12): $a_j^{S*}/a_j^{T*} = (q_j/\xi)^{\eta-1}$, strictly increasing in q_j since $\eta - 1 > 0$ for $\eta > 1$. The service share depends only on bank quality and the CES parameters—not on spreads, the rival's strategy, or rate sensitivity—yielding a clean, directly testable prediction.²⁷ $\square$

Part (iii): $\rho_H^ > \rho_L^*$.* From $\rho_j^* = 1/[\bar{\alpha}_j(1 - D_j)]$, equivalently $\bar{\alpha}_j\rho_j^* = 1/(1 - D_j)$: we need $\bar{\alpha}_H(1 - D_H) < \bar{\alpha}_L(1 - D_L)$. Part (i) gives $\bar{\alpha}_H < \bar{\alpha}_L$ (via (A6)). For the demand side, we prove $D_H > D_L$ by contradiction.

We first establish that $A_H > A_L$. From Part (i), bank H has more total non-price content: $n_H \equiv 1 - a_H^{P*} > 1 - a_L^{P*} \equiv n_L$. At the optimal composition (Part ii), $A_j = n_j \cdot [(q_j s_j^*)^{1-1/\eta} + (\xi(1 - s_j^*))^{1-1/\eta}]^{\eta/(\eta-1)}$ where s_j^* is bank j 's service share within the service-trust split. By the envelope theorem, the bracketed CES value evaluated at the optimal s_j^* is increasing in q_j . Since $n_H > n_L$ and $q_H > q_L$, we have $A_H > A_L$.

Suppose $D_H \leq D_L$. Then $1/(1 - D_H) \leq 1/(1 - D_L)$, i.e., $\bar{\alpha}_H\rho_H^* \leq \bar{\alpha}_L\rho_L^*$. The utility

²⁷In the extended specification with content-specific own-bank effects (Appendix A.3.7, Limitation 2), the composition would additionally depend on the differential attention-diversion advantage of service over trust, making it responsive to market conditions.

difference is:

$$u_H - u_L = (A_H - A_L) + \left[\frac{1}{1 - D_L} - \frac{1}{1 - D_H} \right],$$

where the first term is strictly positive (established above) and the second is non-negative (by the contradiction assumption). Hence $u_H > u_L$, giving $D_H > D_L$ —contradiction. Both channels ($\bar{\alpha}_H < \bar{\alpha}_L$ and $1 - D_H < 1 - D_L$) push $\rho_H^* > \rho_L^*$. $\square$

Existence For fixed content, the rate-setting game is a standard logit pricing game. Each bank's profit is strictly concave in ρ_j when $D_j < 1/2$ (the relevant case with an outside option). The best-response mapping is a contraction under this condition, giving a unique Nash equilibrium in spreads for each content profile. For content, the strategy space is the compact simplex $\{(a_j^P, a_j^S, a_j^T) : a_j^P + a_j^S + a_j^T = 1, a_j^k \geq 0\}$; the CES Inada condition ensures interior (a_j^S, a_j^T) ; the cross-bank externality condition ensures interior a_j^P . The composite payoff function (profit at the equilibrium spread given content) is continuous on the compact content space. By the Debreu-Fan-Glicksberg theorem, a Nash equilibrium in content allocations exists.

Uniqueness. We do not claim global uniqueness of the full content-and-pricing game. Multiple equilibria could in principle arise when quality differentiation is small. Under $q_H/q_L > \theta^*$, the monotone comparative statics argument in Part (i) ensures that *every* equilibrium has $a_H^{P*} < a_L^{P*}$ (and hence $\rho_H^* > \rho_L^*$), because the ranking is determined by dominant-strategy-like logic: Bank H 's best response has lower pricing content than Bank L 's for *all* rival strategies. The equilibrium predictions in Proposition 1 therefore hold in all equilibria, though we cannot rule out multiplicity in the exact values of (a_H^{P*}, a_L^{P*}) . $\square$

A.3.3. Proof of Proposition 2

Non-price content widens spreads. The equilibrium spread satisfies the identity $\bar{\alpha}_j \rho_j^* (1 - D_j) = 1$. To find how ρ_j^* responds to a shift in content, we apply the implicit function theorem (IFT) to this identity, accounting for the feedback through D_j 's dependence on ρ_j^* .

Totally differentiating $\bar{\alpha}_j \rho_j^* (1 - D_j) = 1$ with respect to a_j^S (reducing a_j^P , at fixed a_j^T and

rival strategy):

$$\frac{d\rho_j^*}{da_j^S} \bar{\alpha}_j(1 - D_j) + \rho_j^* \frac{d[\bar{\alpha}_j(1 - D_j)]}{da_j^S} = 0. \quad (\text{A15})$$

The total derivative of $\bar{\alpha}_j(1 - D_j)$ decomposes into a direct effect (through $\bar{\alpha}_j$ and u_j) and a feedback effect (through ρ_j^* 's effect on D_j):

$$\frac{d[\bar{\alpha}_j(1 - D_j)]}{da_j^S} = \underbrace{-\alpha_1(1 - D_j) - \bar{\alpha}_j \frac{\partial D_j}{\partial a_j^S} \Big|_{\rho}}_{\text{direct}} + \underbrace{\bar{\alpha}_j^2 D_j(1 - D_j) \frac{d\rho_j^*}{da_j^S}}_{\text{feedback}},$$

using $\partial D_j / \partial \rho_j = -\bar{\alpha}_j D_j(1 - D_j)$. Substituting into (A15) and collecting terms in $d\rho_j^* / da_j^S$:

$$\frac{d\rho_j^*}{da_j^S} = \frac{\alpha_1(1 - D_j) + \bar{\alpha}_j \frac{\partial D_j}{\partial a_j^S} \Big|_{\rho}}{\bar{\alpha}_j^2(1 - D_j)}. \quad (\text{A16})$$

Evaluating at the content optimum. At the equilibrium content allocation, the content FOC (A14) implies that the bracketed expression in (A13) equals zero, so $\partial D_j / \partial a_j^S \Big|_{\rho} = 0$. Hence:

$$\frac{d\rho_j^*}{da_j^S} \Big|_{a_j^S = a_j^{S*}} = \frac{\alpha_1}{\bar{\alpha}_j^2} > 0. \quad (\text{A17})$$

At the margin, the spread-widening operates entirely through the *rate-sensitivity channel*: each unit of pricing content displaced reduces $\bar{\alpha}_j$ by α_1 , widening the markup. The demand effect is zero at the optimum because the bank has already equated the marginal benefit and cost of non-price content.

Between-bank comparison: the demand channel. While the demand channel vanishes at the margin, it operates through the *level* comparison across banks. In the equilibrium of Proposition 1, Bank H has both lower $\bar{\alpha}_H$ (from Part i) and higher D_H (from Part iii). Both reduce $\bar{\alpha}_j(1 - D_j)$ in the spread formula $\rho_j^* = 1/[\bar{\alpha}_j(1 - D_j)]$, amplifying the spread difference. Thus, the demand channel—while inoperative at the margin—contributes to the equilibrium spread gap $\rho_H^* - \rho_L^*$.

Total spread-widening effect is larger for high-quality banks. From (A17), the marginal spread-widening is $\alpha_1 / \bar{\alpha}_j^2$, which is larger when $\bar{\alpha}_j$ is smaller. Since $\bar{\alpha}_H < \bar{\alpha}_L$ (Proposi-

tion 1(i)):

$$\left. \frac{d\rho_H^*}{da_H^s} \right|_{a_H^{s*}} = \frac{\alpha_1}{\bar{\alpha}_H^2} > \frac{\alpha_1}{\bar{\alpha}_L^2} = \left. \frac{d\rho_L^*}{da_L^s} \right|_{a_L^{s*}}. \quad (\text{A18})$$

The marginal spread-widening is self-reinforcing: banks that have already shifted toward non-price content (low $\bar{\alpha}_j$) gain more from further shifts. This creates a feedback loop that amplifies the equilibrium spread difference between high- and low-quality banks.

Cross-bank spread effect. The rival's non-price content also widens bank j 's spread through the broadcast channel. Holding demand fixed:

$$\left. \frac{\partial \rho_j^*}{\partial a_{-j}^s} \right|_D = \left. \frac{\partial \rho_j^*}{\partial a_{-j}^\tau} \right|_D = \frac{\alpha_2}{\bar{\alpha}_j^2(1 - D_j)} > 0. \quad (\text{A19})$$

When the rival shifts to non-price content, a_{-j}^P falls, reducing $\bar{\alpha}_j$ by α_2 and widening the spread. In general equilibrium, the rival's higher Q_{-j} also reallocates demand, but the rate-sensitivity effect dominates when α_2 is large. $\square$

A.3.4. Deposit Beta Derivation

Micro-foundation: awake and asleep depositors. The effective rate sensitivity $\bar{\alpha}_j$ admits a Bernoulli-draw micro-foundation that connects to the two-period extension. Each depositor evaluating bank j is *awake* to rates with probability λ_j and *asleep* with probability $1 - \lambda_j$. An awake depositor perceives the rate gap with sensitivity $\alpha > 0$; an asleep depositor does not perceive any rate gap. The expected rate sensitivity is $\bar{\alpha}_j = \lambda_j \alpha$, so the wakeup rate is:

$$\lambda_j = \frac{\bar{\alpha}_j}{\alpha} = \frac{\alpha(a_j^P, a_{-j}^P)}{\alpha}, \quad (\text{A20})$$

where $\alpha \geq \sup_{(x,y) \in [0,1]^2} \alpha(x,y)$ ensures $\lambda_j \in [0,1]$ for all feasible content allocations.²⁸ This decomposition is consistent with the Period-1 model: a depositor's expected utility from bank j is $\lambda_j[\alpha(r_j - r_M) + A_j] + (1 - \lambda_j)A_j = \lambda_j\alpha(r_j - r_M) + A_j = -\bar{\alpha}_j\rho_j + A_j = u_j$, exactly as in (3).

²⁸The supremum of $\bar{\alpha}_j$ is attained when pricing content is maximized. For the linear specification, this is $\alpha_0 + \alpha_1 + \alpha_2$ when $a_j^P = 1$ for both banks. Setting α equal to this supremum corresponds to the case where all depositors are awake in the maximum-pricing-content market; any larger α means that even in the most pricing-saturated market, some depositors remain asleep.

Period-2 setup. In Period 2, the policy rate rises to $r'_M = r_M + \Delta r$. Content is pre-committed, so λ_j is inherited from Period 1. Of bank j 's Period-1 depositors (mass D_j): fraction λ_j are awake—they observe the new rate environment and re-optimize via a binary logit with sensitivity α ; fraction $1 - \lambda_j$ are asleep—they do not observe the rate change and remain at bank j regardless. Period-2 deposits:

$$D_j^{(2)}(r_j) = \underbrace{(1 - \lambda_j)D_j}_{\text{asleep: captive}} + \underbrace{\lambda_j D_j \cdot \sigma(r_j - r'_M)}_{\text{awake: re-optimize}}, \quad (\text{A21})$$

where $\sigma(x) \equiv e^{\alpha x} / (1 + e^{\alpha x})$ is the logit retention probability: the probability that an awake depositor stays at bank j given rate gap $x = r_j - r'_M < 0$.

Why logit alone yields $\beta = 1$. Bank j maximizes Period-2 profit $\Pi_j^{(2)} = (r'_M - r_j) \cdot D_j^{(2)}(r_j)$. Let $x \equiv r_j - r'_M$ denote the rate gap (negative when the bank pays less than the market rate). Substituting:

$$\Pi_j^{(2)} = (-x) \cdot D_j [(1 - \lambda_j) + \lambda_j \sigma(x)]. \quad (\text{A22})$$

The Period-1 demand D_j and wakeup rate λ_j are constants (pre-committed). The retention probability $\sigma(x)$ depends only on x . Hence $\Pi_j^{(2)}$ is a function of x alone—it does not depend separately on r'_M . The FOC $d\Pi_j^{(2)}/dx = 0$ pins down a constant optimal gap x^* :

$$(1 - \lambda_j) + \lambda_j \sigma(x^*) = (-x^*) \cdot \lambda_j \alpha \sigma(x^*) (1 - \sigma(x^*)), \quad (\text{A23})$$

using $d\sigma/dx = \alpha \sigma(1 - \sigma)$. Since x^* is independent of r'_M , we have $r_j^* = r'_M + x^*$, so $dr_j^*/dr'_M = 1$ —full pass-through. This translation invariance is a standard property of logit demand: the constant semi-elasticity ensures the optimal markup does not depend on rate levels. A lower wakeup rate λ_j widens the *level* of the spread but does not change the *response* to rate shifts.

Sources of $\beta < 1$. Empirical deposit betas are well below unity (Drechsler et al., 2017), reflecting frictions that break translation invariance: (a) imperfect outside-option pass-through: money market fund yields do not track the policy rate one-for-one, so the effective

outside option rises by less than Δr ; (b) search and adjustment frictions: even awake depositors face information costs and inertia; (c) deposit insurance: FDIC coverage reduces the marginal attractiveness of outside options at high rate levels. Each friction means that a unit increase in r_M translates into less than a unit increase in competitive pressure, breaking the translation invariance that delivers $\beta = 1$.

Bank-specific deposit beta. We adopt the reduced-form assumption that the deposit beta is increasing in $\bar{\alpha}_j$:

$$\beta_j = f(\bar{\alpha}_j), \quad f' > 0, \quad f(0) = 0, \quad \lim_{\bar{\alpha} \rightarrow \infty} f(\bar{\alpha}) = 1. \quad (\text{A24})$$

A tractable parametrization is $\beta_j \approx \bar{\alpha}_j / (\bar{\alpha}_j + \phi)$ with $\phi > 0$ capturing aggregate friction. The monotonicity $f' > 0$ has a clear economic interpretation: banks whose depositors are more awake to rates (higher $\bar{\alpha}_j$, higher λ_j) face faster outflows when lagging the market rate and must pass through more quickly. This monotonicity is shared by all standard friction models and does not depend on the specific functional form.

Example of imperfect outside-option pass-through. Here we provide an example on how imperfect outside-option affects pass-through, as it is the most transparent source and has direct empirical support in Drechsler et al. (2017). Following the empirical evidence that money market fund rates track the policy rate imperfectly, we model the outside option as paying $r_0 = \phi r_M$ in each period, where $\phi \in (0, 1)$ captures the attenuation.

In Period 2, the policy rate rises to $r'_M = r_M + \Delta r$ and the outside option pays $r'_0 = \phi r'_M$. Awake depositors compare the deposit rate r_j against r'_0 . Writing the Period-2 spread as $\rho_j = r'_M - r_j$ and defining $z_j \equiv r_j - \phi r'_M = (1 - \phi)r'_M - \rho_j$, the retention probability for awake depositors is $\sigma(z_j)$. Period-2 profit:

$$\Pi_j^{(2)} = \rho_j D_j \left[(1 - \lambda_j) + \lambda_j \sigma((1 - \phi)r'_M - \rho_j) \right]. \quad (\text{A25})$$

Unlike the $\phi = 1$ case, the optimal ρ_j^* now depends on r'_M through the term $(1 - \phi)r'_M$. This breaks translation invariance: the “effective gap” between the bank and the outside option

shifts with rate levels.

The FOC for ρ_j is:

$$(1 - \lambda_j) + \lambda_j \sigma = \rho_j \lambda_j \alpha \sigma (1 - \sigma), \quad (\text{A26})$$

where $\sigma \equiv \sigma((1 - \phi)r'_M - \rho_j)$. Applying the implicit function theorem to (A26) with respect to r'_M :

$$\beta_j = 1 - \frac{d\rho_j^*}{dr'_M} = \frac{1 + \phi - \phi\rho_j^*\alpha(1 - 2\sigma)}{2 - \rho_j^*\alpha(1 - 2\sigma)}, \quad (\text{A27})$$

where the second-order condition requires $2 - \rho_j^*\alpha(1 - 2\sigma) > 0$. This expression has the key properties:

- *When $\phi = 1$:* $\beta_j = 1$ for all λ_j , recovering full pass-through. The outside option fully tracks the policy rate, restoring translation invariance.
- *When $\phi < 1$:* $\beta_j < 1$. The bank partially absorbs rate increases because its depositors' outside option has not fully adjusted.
- *Monotonicity in λ_j :* As $\lambda_j \rightarrow 0$ (all asleep, $\rho_j^* \rightarrow \infty$, $\sigma \rightarrow 0$): $\beta_j \rightarrow \phi$. As $\lambda_j \rightarrow 1$ (all awake): β_j approaches a value strictly above ϕ and below 1. Since ρ_j^* is decreasing in λ_j (more attentive depositors discipline the spread), and β_j is decreasing in ρ_j^* (wider-spread banks pass through less), β_j is increasing in λ_j .

Since $\lambda_j = \bar{\alpha}_j/\alpha$, β_j is increasing in $\bar{\alpha}_j$. We can write $\beta_j = f(\bar{\alpha}_j)$ with $f' > 0$, $f(0) = \phi$, and $\lim_{\bar{\alpha} \rightarrow \alpha} f(\bar{\alpha}) < 1$, confirming the reduced-form assumption above.

Comparative statics. Since $\partial \bar{\alpha}_j / \partial a_j^S = \partial \bar{\alpha}_j / \partial a_j^T = -\alpha_1 < 0$ (each unit of non-price content displaces one unit of pricing content):

$$\frac{\partial \beta_j}{\partial a_j^S} = \frac{\partial \beta_j}{\partial a_j^T} = f'(\bar{\alpha}_j) \cdot (-\alpha_1) < 0. \quad (\text{A28})$$

Non-price content reduces the bank's deposit beta by putting its depositors to sleep. The effect operates through the own-pricing channel: shifting from pricing to non-price content reduces a_j^P , which lowers $\bar{\alpha}_j$ by α_1 . Through the cross-bank channel, the rival's non-price

content also lowers bank j 's beta via the broadcast spillover: $\partial\beta_j/\partial a_{-j}^s = \partial\beta_j/\partial a_{-j}^T = f'(\bar{\alpha}_j) \cdot (-\alpha_2) < 0$.

Since $\bar{\alpha}_H < \bar{\alpha}_L$ (from $a_H^{P*} < a_L^{P*}$ and Eq. (A6)):

$$\beta_H < \beta_L. \quad (\text{A29})$$

High-quality banks—which compete on non-price dimensions—endogenously have sleepier depositor bases and lower pass-through.

Illustrative parametrization. Under the linear specification $\bar{\alpha}_j = \alpha_0 + \alpha_1 a_j^P + \alpha_2 a_{-j}^P$, suppose the equilibrium content allocations produce $\bar{\alpha}_H \approx 0.10$ and $\bar{\alpha}_L \approx 0.40$, with $\alpha = 0.70$ and $\phi = 0.15$:

- Bank H (service-focused): $\lambda_H = 0.10/0.70 \approx 0.14$, $\beta_H \approx 0.10/(0.10 + 0.15) = 0.40$.
- Bank L (pricing-focused): $\lambda_L = 0.40/0.70 \approx 0.57$, $\beta_L \approx 0.40/(0.40 + 0.15) = 0.73$.

These deposit betas are in the range documented by Drechsler et al. (2017): 0.30–0.40 for sticky banks and 0.60–0.80 for competitive banks. The wakeup rates ($\lambda_H \approx 14\%$, $\lambda_L \approx 57\%$) are broadly consistent with the shopping rates in Egan et al. (2025), who estimate that 5–15% of depositors actively shop for rates each year.

A.3.5. Nonlinear Deposit Dynamics (Prediction for Future Research)

We develop a conjecture about nonlinear deposit dynamics that we flag for future empirical and theoretical work. While our linear model does not generate threshold effects, the mechanism suggests that a richer model with nonlinear attention—where depositors' wakeup probability increases discontinuously when rate gaps exceed a threshold—would predict accelerating outflows for service-focused banks.

The intuition is as follows. Service-focused banks (high $a_j^S + a_j^T$, low $\bar{\alpha}_j$) have mostly asleep depositors who do not notice gradual rate gaps. For small policy shocks, asleep depositors tolerate the widening gap. But when the cumulative shock is large enough, the accumulated rate gap may become salient even to normally inattentive depositors—they “wake up” and defect rapidly.

In pricing-focused markets (high a_j^P , high $\bar{\alpha}_j$), depositors are already awake. Banks adjust rates continuously and depositors respond proportionally, producing approximately linear outflows. This prediction is consistent with the evidence in [Lu and Wu \(2026\)](#) that depositor attention increases when rate gaps are large, and with the 2022–2023 deposit outflow episode in which banks with historically sticky depositor bases experienced sudden, large-scale withdrawals after a rapid tightening cycle.

A.3.6. Welfare Analysis

Non-price advertising has ambiguous effects on depositor welfare. We analyze total surplus, decomposed into depositor surplus and bank profits.

Depositor surplus. In the logit model, expected depositor surplus is $CS = \ln[\exp(u_H) + \exp(u_L) + 1]$ (up to a constant), where $u_j = -\bar{\alpha}_j \rho_j + A_j$. By the logit envelope, $\partial CS / \partial a_j^S = \sum_k D_k \cdot \partial u_k / \partial a_j^S$. Using a chain-rule decomposition that separates the direct effect (holding spreads fixed) from the indirect effect (spread re-optimization):

$$\frac{\partial CS}{\partial a_j^S} = \underbrace{D_j \frac{\partial A_j}{\partial a_j^S}}_{\text{quality gain}} + \underbrace{D_j \alpha_1 \rho_j + D_{-j} \alpha_2 \rho_{-j}}_{\text{direct rate-sensitivity gain}} - \underbrace{\sum_k D_k \bar{\alpha}_k \frac{\partial \rho_k^*}{\partial a_j^S}}_{\text{spread re-optimization cost}}. \quad (\text{A30})$$

The first term is the quality gain from non-price content. The second captures the mechanical benefit of lower rate sensitivity: depositors perceive a smaller rate cost when $\bar{\alpha}_j$ falls, holding spreads fixed.²⁹ The third reflects banks widening spreads in response, partially offsetting the gain.

Bank profits and total surplus. Total bank profit $\Pi = \sum_j \rho_j D_j$ unambiguously increases with non-price content through both channels (lower $\bar{\alpha}_j$ widens spreads; higher A_j raises demand). Total surplus $W = CS + \Pi$ is ambiguous.

²⁹This “gain” reflects depositors becoming less aware of the spread they are paying—it is a gain in perceived utility within the logit framework, not necessarily in objective welfare. Under a welfare standard that weights objective rate costs, this term would be neutral (depositors pay the same spread regardless of attention). We follow the standard logit welfare measure throughout.

Service vs. trust content. The welfare ambiguity is sharper for trust content. Service content generates a genuine quality signal ($\partial A_j / \partial a_j^S$ reflects verifiable quality q_j), improving depositor-bank matching. Trust content generates emotional attachment ($\partial A_j / \partial a_j^T$ reflects ξ) that may not improve matching. This parallels the informative-vs.-persuasive debate in the advertising literature (Bagwell, 2007).

Equilibrium inefficiency. Two sources: (1) a *pricing-content externality*—banks underprovide pricing because they do not internalize the benefit of higher a_j^P to the rival’s depositors (via α_2); (2) a *composition distortion*—banks choose content to maximize own demand rather than aggregate matching quality, potentially over-investing in trust relative to service.

A.3.7. Model Limitations and Extensions

Limitation 1: Boundary conditions. The accommodation equilibrium requires that the profit from the quality-driven demand advantage exceeds the profit from competing on rates. This can fail under small quality differentiation or highly attentive depositor markets.

Limitation 2: Common own-bank effect. The specification treats the own-bank pricing effect α_1 as independent of which non-price content type replaces pricing. An extension with content-specific effects—where shifting to service reduces $\bar{\alpha}_j$ by $\alpha_1 + \gamma \tilde{q}_j$ and shifting to trust reduces it by $\alpha_1 + \delta$, with $\gamma \tilde{q}_H > \delta > \gamma \tilde{q}_L$ —would make high-quality banks’ service advertising more effective at diverting attention, generating additional spread-widening through the rate-sensitivity channel.

Limitation 3: CES specification. The CES aggregation of non-price value is a tractable parametric choice. The key requirements are: (i) interior solutions in both non-price dimensions (Inada condition, satisfied for $\rho < 1$), and (ii) the service share responds to quality (satisfied for $\rho > 0$).

Limitation 4: Exogenous bank quality. Quality q_j is treated as exogenous. Endogenizing quality investment would reinforce the core message: high-quality banks invest in quality and

in service advertising that leverages that quality, creating complementarity between quality and market power.

Limitation 5: Two-bank duopoly. Multibank oligopoly extensions are qualitatively robust under sufficient quality differentiation. The two-bank setup is standard in the deposits channel literature.